Sakartvelo Defenders

Game Design Bible v4.0

Chronologically Accurate Georgian History Tower Defense

10 Eras • 200 Levels • Blockchain-Powered
Godot 4.x • PWA • Immutable zkEVM

This document has been thoroughly revised for historical accuracy. All dates, figures, events, and cultural references have been verified against authoritative academic sources and Georgian historical scholarship. Five factual errors from v3.0 have been corrected, eight nuanced claims have been clarified, and significant previously omitted content has been integrated throughout.

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Confidential Design Document

Table of Contents

1. Vision

2. Game Overview

2.1 Core Gameplay Loop

2.2 Genre and Platform

2.3 Target Audience

3. Eras

3.1 Era 0: Ancient Colchis (~600 BC – ~100 BC)

3.2 Era 1: Kingdom of Iberia (~300 BC – ~630 AD)

3.3 Era 2: Age of Invasions (~630 – ~1089 AD)

3.4 Era 3: The Georgian Golden Age (~1089 – ~1225 AD)

3.5 Era 4: The Mongol Catastrophe (~1225 – ~1500 AD)

3.6 Era 5: Between Empires (~1500 – ~1801 AD)

3.7 Era 6: Russian Empire (~1801 – 1918 AD)

3.8 Era 7: First Democratic Republic (1918 – 1921 AD)

3.9 Era 8: The Soviet Century (1921 – ~1991 AD)

3.10 Era 9: Modern Georgia (~1991 – Present)

4. Tower System

4.1 Tower Upgrade System

4.2 Tower Synergy System

5. Enemies

5.1 Enemy Categories

5.2 Boss Encounters

6. Heroes

6.1 Hero Abilities and Cooldowns

6.2 Hero Progression

7. Maps

7.1 Map Themes by Era

7.2 Elevation Mechanic

8. Tutorial

9. Progression

9.1 Era Unlock Requirements

9.2 Star System

10. Scrolls and Mastery

10.1 Standard Scrolls

10.2 Mastery Scrolls (ERC-721)

10.3 SAKART Boost Scroll

11. Skins and Binding

11.1 Skin Binding

11.2 Marketplace Tooltip

12. Player vs. Player (PvP)

12.1 Attack-Then-Defend Format

12.2 Loadout System

12.3 Defense Setup

12.4 Hero and Scroll Selection

13. Survival Mode

14. Seasons

15. Endgame

16. Educational System

16.1 Cultural Footer

16.2 Historical Accuracy Policy

16.3 Verified Sources and Methodology

17. Mythology Integration

18. SAKART Token Economy

18.1 Token Distribution

18.2 Burn Mechanisms

18.3 Anti-Inflation Design

19. Fees and Revenue Model

19.1 Treasury and Developer Withdrawal

20. Smart Contracts

20.1 SAKART Token Contract (ERC-20)

20.2 Skin Contract (ERC-1155)

20.3 Mastery Scroll Contract (ERC-721)

21. Blockchain Integration

21.1 Wallet Integration

21.2 Chain Outage Fallback

21.3 Invisible Blockchain Philosophy

22. Multi-Account Prevention

23. Competitive Analysis

23.1 Web2 Tower Defense Landscape

23.2 Web3 Gaming Landscape

24. Art Pipeline

24.1 Style Bible System

24.2 AI Generation Workflow

25. Tech Stack

25.1 Godot 4.x and PWA Architecture

25.2 Blockchain Infrastructure

26. Design Principles

27. Roadmap

28. Changelog (v3.0 to v4.0)

1. Vision

Sakartvelo Defenders is a historically accurate tower defense game set across the full sweep of Georgian history, from the ancient Kingdom of Colchis to the modern Republic of Georgia. The game is built on three foundational pillars: education through gameplay, invisible blockchain integration, and cultural preservation. Players defend Georgian lands against historically authentic waves of invaders across ten distinct eras, each meticulously researched and visually distinct.

The core philosophy is that players should never feel like they are interacting with a blockchain product. All Web3 elements, including the SAKART token economy, NFT-based mastery scrolls, and tradable cosmetic skins, operate beneath the surface. The gameplay experience is indistinguishable from a premium Web2 tower defense title, with blockchain providing true asset ownership and player-driven economies as invisible infrastructure rather than visible friction.

Georgia's history is one of the most dramatic and underrepresented narratives in gaming. Positioned at the crossroads of Europe and Asia, Georgia has faced invasions from virtually every regional power: Persians, Romans, Arabs, Mongols, Ottomans, Persians again, and Russians. Each era produced unique military technologies, heroic figures, and cultural achievements that translate naturally into compelling tower defense mechanics. The game's ten-era structure ensures players experience this entire arc, learning Georgian history organically through hundreds of levels of strategic gameplay.

Historical accuracy is not merely a design preference but a core commitment. Every date, figure, event, and cultural detail depicted in this document has been cross-checked against authoritative academic sources, Georgian historical records, and peer-reviewed scholarship. Where the historical record is ambiguous or contested, the document acknowledges the debate rather than presenting one interpretation as settled fact. This commitment extends from the broad era structure down to individual hero abilities, enemy designs, and educational pop-up content. Inaccurate history would undermine the game's educational mission and disrespect the culture it aims to celebrate.

2. Game Overview

2.1 Core Gameplay Loop

The fundamental gameplay loop of Sakartvelo Defenders follows the classic tower defense formula, refined with historical authenticity and strategic depth. Players progress through levels by placing defensive towers along enemy approach paths, utilizing heroes with unique abilities, and activating consumable scrolls at critical moments. Each level presents a map inspired by real Georgian landscapes and fortifications, with enemy waves that reflect the actual historical invaders of each era.

The loop begins with level selection within an era. Players assess the map layout, choose their tower composition from their available roster, and place towers strategically during a build phase. Enemy waves then advance along predetermined paths, and players must manage their economy by earning resources from defeated enemies to build additional towers or upgrade existing ones. Heroes provide active abilities that players trigger manually, adding a layer of real-time decision-making. Scrolls offer powerful one-time effects that can turn the tide of difficult engagements.

Completion is measured through a three-star system. Each star represents a progressively harder challenge tier within the same level: surviving all waves (1 star), completing with a life threshold (2 stars), and completing under a strict time or resource constraint (3 stars). Each star earns a one-time SAKART token reward, establishing a natural economy where progression drives token distribution without allowing replay farming.

2.2 Genre and Platform

Sakartvelo Defenders is a 2D isometric tower defense game. The isometric perspective provides clear visibility of enemy paths, tower ranges, and strategic positioning while maintaining the aesthetic warmth of illustrated historical environments. The 2D approach enables efficient asset production using AI-generated art with strict style bible consistency controls, making the project feasible for a solo developer operating on a minimal budget.

The game is developed in Godot 4.x and deployed as a Progressive Web App (PWA). This architectural choice eliminates server hosting costs entirely, as the game runs client-side in any modern browser. The PWA wrapper enables offline gameplay for the full campaign, with blockchain transactions queued locally

during chain outages and submitted automatically when connectivity resumes. This ensures the player experience is never interrupted by infrastructure issues.

2.3 Target Audience

The primary audience encompasses tower defense enthusiasts, Web3 gaming early adopters, and players interested in history and cultural education. Tower defense fans will appreciate the strategic depth of synergy combinations, hero ability management, and the challenge of three-star completion across hundreds of levels. Web3 players will find genuine utility in the SAKART token economy without the usual crypto-game friction of mandatory wallet connections or pay-to-win mechanics.

A secondary but equally important audience is the Georgian diaspora community and cultural preservation advocates. Georgia has a rich cultural heritage that is underrepresented in mainstream gaming. By presenting this history through an accessible, engaging format, the game serves as both entertainment and educational tool. Historical pop-ups appear between levels, offering context about real events, figures, and locations depicted in the gameplay.

A tertiary audience includes educators and institutions. The game's detailed historical content, verified against academic sources, makes it suitable for classroom use as a supplementary educational tool. The educational system (Section 16) is designed to be informative enough for formal learning contexts while remaining engaging enough to hold student attention.

3. Eras

The game spans ten chronological eras, each representing a major period in Georgian history. Every era contains 20 levels, totaling 200 levels across the full campaign. The era structure follows a strict chronological progression, ensuring historical accuracy and allowing players to experience the complete narrative arc of Georgian civilization. Each era introduces new tower types, enemy variants, hero characters, and synergy combinations, providing continuous mechanical evolution alongside historical progression.

The chronological approach is non-negotiable. Georgian history is a continuous story where events in one era directly shape the next. The Arab invasions of the 7th century created the conditions for David IV's reforms in the 11th century. The Great Turkish Onslaught (Didi Turkoba) of the 1080s devastated Georgia so thoroughly that David's reconquest seemed miraculous. The Mongol devastation of the 13th century led directly to the fragmentation that made Georgia vulnerable to Ottoman and Persian domination. By maintaining strict

chronological order, the game preserves historical cause-and-effect relationships that would be lost with a thematic or arbitrary ordering.

Era	Name	Period	Levels	Key Figures
0	Ancient Colchis	~600 BC – ~100 BC	20	Medea, Aeëtes, Jason
1	Kingdom of Iberia	~300 BC – ~630 AD	20	Pharnavaz, St. Nino, Vakhtang Gorgasali
2	Age of Invasions	~630 – ~1089 AD	20	Bagrat III, Arab Emirs, Bagrationi Kings
3	The Georgian Golden Age	~1089 – ~1225 AD	20	David IV, Queen Tamar, Zakare Zakarian
4	The Mongol Catastrophe	~1225 – ~1500 AD	20	Queen Rusudan, George V, Mongol Generals
5	Between Empires	~1500 – ~1801 AD	20	Erekle II, Vakhtang VI, Persian/ Ottoman Rivals
6	Russian Empire	1801 – 1918 AD	20	Pyotr Bagration, Ilia Chavchavadze
7	First Democratic Republic	1918 – 1921 AD	20	Noe Jordania, Independence Fighters
8	The Soviet Century	1921 – ~1991 AD	20	Stalin (historical context), Dissident Figures
9	Modern Georgia	~1991 – Present	20	Gamsakhurdia, Saakashvili

3.1 Era 0: Ancient Colchis (~600 BC – ~100 BC)

> **v4.0 Correction:** The date range has been changed from ~1500 BC – ~300 BC to ~600 BC – ~100 BC. The earlier range included pre-kingdom Bronze Age settlements. The Kingdom of Colchis as an organized polity dates to approximately the 6th century BC, when Colchian tribes coalesced into a recognizable kingdom under Greek historical record. The endpoint has been adjusted to reflect the conquest of Colchis by Mithridates VI of Pontus around 83 BC, after which Colchis lost its independence and became a Roman client territory.

The opening era of Sakartvelo Defenders transports players to the ancient Kingdom of Colchis, the land of the Golden Fleece. This era draws from Greek mythological sources, archaeological evidence, and the limited but significant historical records of Colchian civilization. Colchis was a wealthy kingdom located

on the eastern shore of the Black Sea, known for its gold mining, metalworking, and advanced agricultural practices. The Greek myth of Jason and the Argonauts reflects the historical reality of Greek interest in Colchian gold and trade routes along the Phasis (modern Rioni) River. Colchis was also notable for its independent development of iron metallurgy and its sophisticated gold-smithing techniques, which produced some of the finest metalwork of the ancient world.

The Kingdom of Colchis occupied what is now western Georgia, encompassing the fertile lowlands along the Black Sea coast and the river valleys that cut into the Caucasus foothills. The Colchians were ethnically distinct from their Iberian neighbors to the east, speaking a language related to modern Mingrelian and Laz (the Zan branch of the Kartvelian language family). Colchian society was organized around a powerful monarchy centered on the city of Aea (possibly near modern Kutaisi), with a priestly class that maintained temples to native deities. The kingdom's wealth came from its gold deposits, its strategic position on Black Sea trade routes, and its agricultural abundance, particularly in linen, hemp, and timber.

Colchis maintained complex relationships with the Greek world. Greek colonies such as Phasis and Dioscurias (modern Sukhumi) were established along the Colchian coast from the 6th century BC, creating trade networks that brought Greek goods, ideas, and military pressure into Colchian territory. The kingdom also came under the nominal suzerainty of the Achaemenid Persian Empire, though Colchian autonomy was largely preserved. By the 1st century BC, Colchis was absorbed into the Kingdom of Pontus under Mithridates VI, and subsequently fell under Roman influence after Pompey's campaigns in the Caucasus (65 BC). The kingdom effectively ceased to exist as an independent polity by approximately 83 BC, though Colchian cultural identity persisted for centuries.

Gameplay in this era introduces fundamental tower defense mechanics with a primitive, bronze-age aesthetic. Towers are stylized as wooden watchtowers, stone fortifications, and archer platforms reflecting the military technology of the period. Enemy waves represent tribal raiders from the mountains, early Greek expeditionary forces seeking Colchian gold, and mythological creatures drawn from Georgian folklore such as the devi (many-headed ogres whose heads regenerate if not completely destroyed, comparable to the Lernaean Hydra). The environment features lush Colchian forests, river valleys of the Rioni, the ancient city of Vani (one of the most important and best-studied archaeological sites of Colchian civilization, inscribed on UNESCO's tentative World Heritage list), and the Greek colonial ports along the coast.

The hero for this era is Medea, the Colchian princess from Greek mythology. In modern Georgian culture, Medea is celebrated as a symbol of Colchian civilization and its advanced knowledge of medicine, herbalism, and metalworking. The prominent Medea statue in Batumi stands as a national

monument to Georgia's ancient heritage, depicting her holding the Golden Fleece. While Greek sources cast Medea as a complex and ultimately tragic figure who helped Jason steal the Fleece and then turned to vengeance when he abandoned her, Georgian cultural interpretation emphasizes her identity as a powerful Colchian princess, healer, and possessor of arcane knowledge. Her abilities reflect her mythological association with magic and herbalism, providing healing buffs for nearby towers and poison damage to enemies. Her ultimate ability, "Colchian Alchemy," temporarily transforms all nearby towers into enhanced versions with special effects, representing the transformative power of Colchian knowledge.

The educational pop-ups for this era cover the archaeological evidence for Colchian civilization at Vani and other sites, the reality behind the Golden Fleece legend (widely believed to reference the ancient technique of gold-panning using sheep fleeces to trap gold particles in mountain streams, a method still practiced in Svaneti into the 20th century), the cultural exchange between Colchis and the Greek world, and the distinct ethnic and linguistic identity of the Colchian people as ancestors of modern western Georgians.

3.2 Era 1: Kingdom of Iberia (~300 BC – ~630 AD)

The Kingdom of Iberia (Kartli in Georgian) represents the formation of the Georgian state as a major regional power. This era spans over nine centuries, from the establishment of the Iberian kingdom under King Pharnavaz around 299 BC to the Arab incursions of the 7th century AD that would fundamentally reshape Georgian political life. It encompasses the Roman-Parthian rivalry for control of the Caucasus, the Christianization of Georgia by Saint Nino in the early 4th century AD, and the reign of King Vakhtang Gorgasali, who established Tbilisi as the capital.

King Pharnavaz I (r. c. 299–234 BC) is regarded as the founder of the Iberian kingdom and the progenitor of the Pharnavazid dynasty. According to the medieval Georgian chronicle "Kartlis Tskhovreba" (Life of Kartli), Pharnavaz unified the Georgian tribes, established a centralized state, created an administrative division of the kingdom into saeristavos (provinces governed by eristavis), and initiated the construction of the fortress of Armazi near Mtskheta. He also established the worship of the deity Armazi (from whom the fortress took its name), reflecting the pre-Christian religious practices of ancient Iberia. While the chronicles attribute these achievements to Pharnavaz, modern scholars note that some elements may reflect later projections onto an ancestral figure, though the broad outlines of his reign are accepted as historically grounded.

Pre-Christian Iberian religion was a syncretic system combining indigenous Georgian deities with Iranian and Anatolian influences. The supreme deity was

Armazi, a sky/thunder god whose idol was erected at the fortress of Armazi near Mtskheta. Other major deities included Zaden (a fertility god), Gatsi (associated with the moon), and Gaim (associated with the sun). The worship of these deities was centered on the Armazi fortress and the pagan shrines of Mtskheta. The conversion of Iberia to Christianity in the 4th century led to the systematic destruction of these pagan sites, though elements of pre-Christian belief persisted in folk traditions for centuries. The educational content for this era should reference these deities, as they provide important cultural context and could be incorporated into tower names, ability descriptions, and shrine mechanics.

Saint Nino of Cappadocia brought Christianity to Iberia in the early 4th century AD, traditionally dated to approximately 327–337 AD during the reign of King Mirian III. According to Georgian tradition, Nino was a Roman woman who traveled to Iberia bearing a cross made of vine branches bound with her own hair. Through her preaching and reported miracles, she converted first the queen, Nana, and then King Mirian himself, who declared Christianity the state religion of Iberia. This made Iberia one of the earliest Christian kingdoms in the world, predating the Roman Empire's official adoption of Christianity. The conversion was not merely a religious event but a geopolitical reorientation, aligning Iberia with the Christian Roman/Byzantine world against Zoroastrian Persia. The consequences of this alignment would shape Georgian history for centuries, as Georgia became a frontline state in the struggle between Christian and Islamic civilizations.

King Vakhtang I Gorgasali (r. c. 447–522 AD) is one of the most celebrated figures in Georgian history. His sobriquet "Gorgasali" derives from the Persian "Gorg-aslan," meaning "wolf-head," reportedly given because he wore a wolf's head helmet into battle and fought with savage ferocity. Vakhtang is traditionally credited with establishing Tbilisi as the capital of Iberia. While the area of Tbilisi was already inhabited before his reign, with archaeological evidence suggesting settlement dating back to at least the 4th century AD, Vakhtang significantly developed the existing settlement around its famous hot springs (the name "Tbilisi" derives from the Georgian "tbili," meaning warm) and moved the seat of government there from Mtskheta. The famous legend tells of Vakhtang discovering the hot springs while hunting, when his falcon caught a pheasant that fell into the boiling springs and was cooked. Impressed by the thermal waters, the king ordered a city built on the site.

Vakhtang Gorgasali's reign was marked by ambitious military campaigns against both the Persians and the Byzantines, attempting to assert Iberian independence between the two great powers. His wars against Persia ultimately proved fatal; he died in battle against the Sassanid Persian army in 522 AD. He is venerated as a saint in the Georgian Orthodox Church, and his legacy as a warrior-king who defied empires resonates deeply in Georgian national identity.

Towers in this era evolve from simple wooden structures to stone fortifications and early siege equipment, reflecting the technological progression of the period. New enemy types include Roman legionaries (representing Rome's periodic military interventions in the Caucasus), Parthian cavalry, Sassanid Persian forces, and early Arab raiders. The introduction of Christianity as the state religion provides narrative context for the era, with churches and monasteries appearing as defendable structures on maps. Environment art shifts to reflect the growing urbanization of Iberia, with maps set in the ancient capital of Mtskheta, the cave city of Uplistsikhe (one of the oldest known settlements in Georgia, carved into the rock and continuously inhabited from the early Iron Age), the fortress of Armazi, and the newly established city of Tbilisi.

Three heroes are available in this era: King Pharnavaz (founder of the kingdom, provides kingdom-wide buff abilities that boost tower efficiency and reduce placement costs, reflecting his historical reputation as a state-builder and organizer), Saint Nino (healer and protector, with abilities that shield towers and create slowing auras around enemy paths, reflecting her role as a spiritual protector of the Georgian people), and King Vakhtang Gorgasali (military commander with aggressive offensive abilities that boost tower damage in an area and a charge ability that damages enemies near his position, reflecting his reputation as a fearsome warrior-king). The era concludes with the devastating Arab invasions that will set the stage for the next era.

3.3 Era 2: Age of Invasions (~630 – ~1089 AD)

The Age of Invasions covers nearly five centuries of continuous foreign pressure on Georgian lands, a period that tested Georgian civilization to its limits and ultimately forged the national identity that would produce the Golden Age. This era encompasses the Arab occupation, the slow reconquest by the Bagrationi dynasty, and the devastating Great Turkish Onslaught (Didi Turkoba) that nearly destroyed Georgia before David IV's miraculous restoration.

Arab forces first reached the Caucasus in the 640s AD. The first Arab incursion into Tbilisi occurred around 645 AD, when Arab forces captured the city during the early expansion of the Rashidun Caliphate. However, this initial conquest did not immediately establish stable Arab control. The Emirate of Tbilisi was not formally established until 736 CE under the Umayyad Caliphate, when Marwan ibn Muhammad (who would later become the last Umayyad caliph) consolidated Arab administrative and military control over the region. The Tbilisi Emirate became a significant center of Islamic power in the Caucasus, ruling the city and its surrounding territories for centuries. The emirs of Tbilisi were powerful figures who at times operated with near-independence from the caliphate, at times acknowledging Abbasid authority, and at times pursuing their own regional ambitions.

> **v4.0 Correction:** The v3.0 Bible conflated the first Arab capture of Tbilisi (c. 645 AD) with the formal establishment of the Tbilisi Emirate (736 CE). These are two distinct events separated by nearly a century. The period between 645 and 736 was characterized by Arab raids, fluctuating control, and Georgian resistance, not a stable emirate.

During the Arab period, Georgian political life fragmented. The ruling house retreated to the southwestern provinces of Tao-Klarjeti, where the Bagrationi (Bagratid) dynasty slowly rebuilt its power base. The Bagrationi dynasty, which would eventually unify Georgia and produce its greatest kings, likely originated from the Armenian Bagratuni dynasty, though Georgian tradition traces their lineage to the biblical King David. The family established itself in Tao-Klarjeti in the 8th century and began the slow process of reconquest, gradually reclaiming territories from the Arab emirs and re-establishing Georgian Christian governance.

The crucial unifying figure of this era was King Bagrat III (r. 1008–1014), who accomplished what no Georgian ruler had achieved in centuries: the unification of Georgia. Bagrat inherited the kingdom of Abkhazia through his mother and the kingdom of Kartli through his father, and in 1008 he was crowned as the first ruler of a unified Georgian kingdom encompassing both western and eastern Georgia. The unification was not merely a personal union but a genuine political consolidation that laid the foundation for the Georgian Golden Age. Bagrat III's achievement was directly responsible for the centralized state that David IV would later transform into a regional superpower.

The era also saw significant military conflicts with the Byzantine Empire, particularly the Battle of Shirimni (September 11, 1021), where Georgian King George I fought the Byzantine Emperor Basil II over control of the disputed territory of Tao-Klarjeti, and the Battle of Sasireti (1042), where King Bagrat IV was defeated by the powerful rebel feudal lord Liparit IV of Kldekari during a Georgian civil war. These conflicts demonstrate the complex political environment of the period, where Georgia faced threats not only from Arab powers but also from Byzantine encroachment and internal feudal fragmentation.

The most devastating event of this era was the Great Turkish Onslaught (Didi Turkoba) of the 1080s. Massive Seljuk Turk invasions swept through Georgia, devastating the countryside, depopulating entire regions, and reducing the Georgian kingdom to a shadow of its former self. The Turkish nomads who flooded into the Georgian heartland were not organized armies but large tribal migrations that destroyed agricultural infrastructure, disrupted trade networks, and forced the Georgian population to retreat into fortified mountain strongholds. Georgian chronicles describe the land as laid waste, with churches destroyed, villages abandoned, and the king reduced to a fugitive in his own kingdom. The Didi Turkoba is essential context for understanding the Golden Age

that follows: David IV's achievements were so remarkable precisely because he inherited a nearly destroyed kingdom and rebuilt it from the ruins.

Mechanically, this era introduces the concept of fortified positions and defensive sieges. Maps feature crumbling Iberian fortresses being held against numerically superior Arab and Seljuk forces. New tower types include fortified gatehouses and mounted archer platforms. Enemy waves are larger and more aggressive, reflecting the overwhelming military pressure of the period. The difficulty curve increases significantly, as players must defend against sustained multi-wave assaults from enemies with greater variety and strength. The later levels of this era introduce the Didi Turkoba mechanic, where enemy waves are dramatically larger and more varied, simulating the overwhelming pressure of the Seljuk migrations.

The heroes of this era represent the resistance and resilience of the Georgian people during centuries of foreign domination. King Bagrat III the Unifier provides abilities centered on consolidation and unity: his "Unification" ability temporarily boosts all tower stats, reflecting his historical achievement of bringing the Georgian lands together. His passive bonus reduces the cost of tower placement in different sectors of the map, representing the economic benefits of a unified state. Bagrationi princes who led the slow reconquest feature as additional heroes, alongside religious figures who maintained Georgian cultural identity during the occupation.

The educational content covers the Tbilisi Emirate and the distinction between the 645 AD Arab conquest and the 736 CE formal establishment of the emirate, the Bagrationi dynasty's origins and their gradual reconquest from Tao-Klarjeti, the significance of Bagrat III's unification, the Battle of Shirimni, the Battle of Sasireti, the Didi Turkoba and its devastating impact, and the broader geopolitical context of Arab-Byzantine-Seljuk rivalry in the Caucasus.

3.4 Era 3: The Georgian Golden Age (~1089 – ~1225 AD)

The Georgian Golden Age is the centerpiece of the game, representing the period when Georgia reached its greatest territorial extent, cultural flowering, and military power. Initiated by King David IV (David the Builder, r. 1089–1125), who reformed the military, reunited the Georgian territories, and defeated the Seljuk Turks at the Battle of Didgori on August 12, 1121, this era culminated under Queen Tamar (r. 1184–1213), during whose reign Georgia controlled territories from the Black Sea to the Caspian and from the Caucasus Mountains to northern Iran and Anatolia. The Golden Age represents not only military and political power but also an extraordinary cultural renaissance that produced some of the finest art, architecture, and literature of the medieval Christian world.

David IV inherited a kingdom on the verge of collapse. The Didi Turkoba had devastated Georgia, the nobility was fractured, and the Seljuk Turks controlled much of the countryside. David's response was systematic and far-reaching. He reformed the military by creating the monaspa, a professional standing army of hired soldiers (the term derives from Georgian "mona" meaning hired and "spa" meaning army), which reduced the kingdom's dependence on unreliable feudal levies. He settled 40,000 Kipchak (Cuman) tribal families from the northern Caucasus steppes in Georgia, providing them with land in exchange for military service; these Kipchak warriors became the backbone of the revitalized Georgian army and one of the most feared cavalry forces in the region. He rebuilt cities, restored churches, and re-established centralized royal authority over the fractious nobility.

The decisive moment of David's reign was the Battle of Didgori on August 12, 1121. A massive coalition of Seljuk Turkish forces, reportedly numbering in the hundreds of thousands according to Georgian chronicles (though modern estimates suggest a more conservative figure of perhaps 30,000–60,000 for the Muslim coalition), invaded Georgia to crush David's resurgent kingdom. David, commanding a significantly smaller force, employed brilliant tactical deception, dividing his army into smaller units that attacked the Turkish camp from multiple directions simultaneously. The confusion in the Turkish ranks turned into a rout, and the Georgian victory was so complete that the day became known as "Dzleva Sakvirveli" (the Wonderful Victory). Didgori is commemorated annually in Georgia and is considered one of the greatest military victories in the nation's history. The battle opened the way for the reconquest of Tbilisi, which David liberated from Arab rule in 1122 after nearly four centuries of Muslim control.

> **Historical Nuance:** The numbers at Didgori are disputed. Georgian chronicles claim the Muslim army numbered 100,000 to 250,000 or even 600,000, while Arab and Turkish sources suggest figures closer to 30,000. Modern historians generally agree that the Georgian sources exaggerate the enemy numbers to enhance the miracle of the victory. The game's educational content should present both the traditional Georgian figures and the modern scholarly estimates, noting that regardless of the exact numbers, the Georgian victory was remarkable and decisive.

Queen Tamar's reign represents the absolute apex of Georgian power and culture. Tamar was crowned as "King of Kings" (mepe ta mepe in Georgian), ruling as a monarch in her own right rather than as a consort—a radical concept in medieval Europe and the Islamic world alike. Her reign saw Georgian armies campaign successfully from the Black Sea coast to the Caspian, from the Caucasus passes to the plains of northern Iran. Georgian forces under the command of the Zakarian brothers conquered vast territories in the Caucasus and eastern Anatolia, establishing Georgian influence over a network of client states and tributary peoples.

Tamar's personal life was as dramatic as her reign was glorious. Her first marriage, arranged by the powerful court faction led by the Catholicos and the amirspasalar (commander-in-chief), was to Yuri Bogolyubsky, a Rus prince from Vladimir-Suzdal. This marriage was a political disaster. Yuri proved dissolute, violent, and ambitious, alienating the Georgian court and nobility. After he attempted to seize power and raised a rebellion against Tamar, the marriage was annulled around 1187 and Yuri was expelled from Georgia (he would later return with a rebellion force but was defeated). Tamar's second marriage, to David Soslan, an Alan (Ossetian) prince and Bagrationi descendant through a secondary line, was far more successful. David Soslan proved a capable military commander who led Georgian armies in Tamar's name, winning major victories at Shamkor (1195) and Basian (1203). Their children, George IV Lasha and Rusudan, would succeed Tamar on the Georgian throne.

> **v4.0 Addition:** Tamar's two marriages are a significant omission from v3.0. Both marriages are historically important: the first demonstrates the political pressures on a female ruler and the consequences of a poor political marriage, while the second shows a successful partnership that sustained the Golden Age. David Soslan's military leadership was essential to Tamar's conquests.

The Zakarian brothers, Zakare and Ivane, were the principal military commanders of Tamar's reign and the architects of Georgia's territorial expansion. The brothers were Armenian princes of the Zakarian (also known as Mkhargrdzeli in Georgian, meaning "long-eared," likely a sobriquet rather than a literal description) family who entered Georgian service. Ivane later converted to Georgian Orthodox Christianity, while Zakare remained Armenian Apostolic in faith. Together, they commanded the combined Armenian-Georgian armies that conquered Ani, Dvin, and other major cities of the Caucasus and eastern Anatolia. Their military campaigns extended Georgian control from the Black Sea to the Caspian, making Georgia the dominant power in the region. The Zakarian conquests are a testament to the multicultural nature of the Georgian Golden Age, where Armenian generals could rise to the highest military positions in service to a Georgian monarch.

> **v4.0 Correction:** The v3.0 Bible referred to "Zakaria Mkhargrdzeli," which uses the Georgian sobriquet rather than the standard academic name. The family is universally known in English-language scholarship as the "Zakarian" brothers (or "Zakarid" dynasty). "Mkhargrdzeli" is the Georgian form, but "Zakarian" is the primary name in academic literature. Additionally, the v3.0 Bible did not acknowledge their Armenian origin, which is a well-documented and significant aspect of their identity. The hero in this Bible uses "Zakare Zakarian" as the primary name, with "Mkhargrdzeli" noted as the Georgian alternative.

Another significant dimension of Tamar's foreign policy was the establishment of the Empire of Trebizond as a Georgian client state. In 1204, following the Fourth

Crusade's sack of Constantinople, Tamar provided military and financial support to Alexios and David Komnenos, grandsons of the Byzantine emperor Andronikos I, enabling them to establish the Empire of Trebizond on the southern coast of the Black Sea. Trebizond served as a Georgian-friendly buffer state and trading partner, demonstrating the reach and sophistication of Georgian diplomacy during the Golden Age.

This era features the most diverse tower roster in the game. Advanced fortifications, including the massive stone castles that dot the Georgian countryside, are joined by specialized units reflecting the professionalized Georgian military of the period. Monaspa royal guard towers, Kipchak cavalry platforms, and siege defense emplacements provide players with powerful defensive options. Enemy waves represent the various peoples who challenged Georgian hegemony: remnants of Seljuk forces, Turkmen tribes, and northern Caucasian mountaineers.

Both David IV and Queen Tamar are playable heroes in this era, as both figures are essential to the period. David IV provides military leadership buffs that increase tower damage and reduce upgrade costs, reflecting his historical reputation as the great reformer who rebuilt Georgia from ruin. His "Dzleva Sakvirveli" ultimate ability temporarily doubles all tower damage for a short duration, representing the miraculous victory at Didgori. Queen Tamar offers diplomatic and governance abilities that generate bonus resources and provide temporary invulnerability to structures, reflecting her reign of prosperity. Her "King of Kings" ultimate ability makes all towers on the map temporarily invulnerable and maximizes their damage output, representing the apex of Georgian power under her rule.

Additional heroes from this era include the poet Shota Rustaveli, whose epic poem "The Knight in the Panther's Skin" (Vepkhistqaosani) is the national epic of Georgia and one of the masterpieces of medieval world literature. Rustaveli's hero ability is "Cultural Inspiration," a passive bonus that increases the attack speed and damage of all towers on the map by a modest percentage, reflecting the cultural flowering that accompanied and sustained Georgia's military power rather than direct combat ability. The identity of the animal in the poem's title is debated among scholars: in modern Georgian, the word refers to a tiger, but in Rustaveli's time it could have meant panther or leopard. "The Knight in the Panther's Skin" is the most widely used English translation and the one adopted by the game. The military commander Zakare Zakarian provides offensive abilities that boost cavalry-type towers and enable coordinated strikes against enemy formations.

The educational content for this era covers the monaspa reform and Kipchak integration, the Battle of Didgori and the liberation of Tbilisi, Tamar's reign and her two marriages, the Zakarian brothers and their Armenian origin, the Empire

of Trebizond as a Georgian client state, the cultural achievements of the Golden Age including Rustaveli's epic and the construction of Gelati Monastery and Academy, and the architectural legacy visible in churches, fortresses, and cave cities like Vardzia.

3.5 Era 4: The Mongol Catastrophe (~1225 – ~1500 AD)

The Mongol Catastrophe marks the abrupt end of the Georgian Golden Age and the beginning of centuries of decline. Beginning with the first Mongol reconnaissance into the Caucasus in 1220, when generals Subutai and Jebe passed through Georgian territory while pursuing the Khwarezmian Shah, and culminating in the complete subjugation of Georgia by 1243, this era represents the most devastating period in Georgian history. The Mongol conquest destroyed the centralized Georgian state, decimated the population, and ushered in nearly three centuries of fragmentation and decline.

The first Mongol contact with Georgia came in 1220, when Subutai and Jebe's reconnaissance force passed through the Caucasus during their extraordinary cavalry raid across the former Khwarezmian Empire. The Georgians under Ivane I Zakarian attempted to block the Mongol passage but were defeated. However, this was merely a reconnaissance, not a conquest. The full-scale Mongol conquest of Georgia began in 1236 under General Chormaqan, who systematically subjugated the Caucasus over the following years. By 1243, Georgia had been forced to submit to Mongol overlordship, paying heavy tribute and providing troops for Mongol campaigns. The kingdom was divided between the two Mongol successor states: the Golden Horde in the northwest and the Ilkhanate in the southeast, each demanding tribute and military service.

Queen Rusudan (r. 1223–1245), Tamar's daughter and successor, was the last ruler of the unified Georgian Golden Age state. She inherited a kingdom still powerful but facing unprecedented threats. The Mongol invasion forced her to flee Tbilisi and seek refuge in Kutaisi. Her reign ended with Georgia's submission to Mongol authority, a bitter conclusion to the era of Georgian greatness. Rusudan's attempts to negotiate with the Pope and the European crusaders for assistance against the Mongols came to nothing, and Georgia was left to face the Mongol storm alone.

George V the Brilliant (r. 1299–1302 and 1314–1346) is one of the few bright spots in this dark era. He ruled twice: first briefly from 1299 to 1302 during a period of Mongol-installed puppet rulers, and then again from 1314 to 1346, during which he accomplished the remarkable feat of reuniting Georgia and briefly throwing off Mongol overlordship. George V used the decline of the Ilkhanate to his advantage, skillfully exploiting Mongol internal conflicts to

restore Georgian independence. His reign saw the restoration of the Georgian Orthodox Church's independence and the rebuilding of churches and monasteries that had been destroyed during the Mongol period. However, his reunification proved temporary; after his death, Georgia fragmented once again into competing kingdoms and principalities.

> **Historical Nuance:** George V had two separate reigns (1299–1302 and 1314–1346), not a single continuous reign. This is an important detail that the game should represent accurately, as it demonstrates the turbulence of the period.

The final blow to Georgian power came from Timur (Tamerlane), who invaded Georgia eight times between 1386 and 1403 with devastating consequences. Timur's invasions destroyed what little remained of Georgia's infrastructure, depopulated entire regions, and ensured that the kingdom would never recover its former unity. The Timurid invasions were so destructive that some Georgian regions did not recover their pre-Mongol population levels until the 18th century.

Gameplay in this era shifts dramatically in tone. Players are no longer defending a unified, prosperous kingdom but rather struggling to survive overwhelming odds with limited resources. Tower placement is constrained by crumbling infrastructure, and enemy waves are massive Mongol armies with superior mobility and firepower. New mechanics introduced in this era include retreat and evacuation objectives, where players must protect civilian populations as they withdraw to mountain fortresses. The environment art reflects the devastation: burned cities, abandoned villages, and fortified mountain redoubts. A brief mid-era narrative arc featuring George V the Brilliant's reunification provides a temporary respite, with gameplay mechanics that briefly mirror the Golden Age before the Timurid invasions plunge the era back into darkness.

The hero for this era represents the last generation of unified Georgian resistance. Queen Rusudan provides diplomatic abilities that generate emergency resources and temporarily reduce enemy damage, reflecting her desperate attempts to find allies against the Mongols. George V the Brilliant provides a "Restoration" ultimate ability that temporarily repairs all damaged towers and boosts their stats, representing his brief reunification of the kingdom. The boss encounters for this era feature Mongol General Chormaqan (who led the 1236–1243 conquest) and Timur himself in the era's climactic battle.

The educational content covers the Mongol administrative system in Georgia, the economic exploitation that drained the kingdom's wealth, Queen Rusudan's failed diplomatic appeals to the Pope and European powers, George V's two reigns and his brief restoration of independence, Timur's devastating invasions, and the cultural resilience that kept Georgian identity alive even under foreign domination. This era sets the stage for the fragmentation that defines the subsequent era.

3.6 Era 5: Between Empires (~1500 – ~1801 AD)

Between Empires covers the period of Georgian history when the fragmented Georgian kingdoms found themselves caught between the Ottoman Empire and Safavid (later Afsharid and Qajar) Persia. This era saw both devastating wars and remarkable cultural achievements, including the reign of Vakhtang VI the Lawgiver and the heroic but ultimately tragic efforts of King Erekle II to maintain Georgian independence. Despite his efforts, including a disastrous alliance with Catherine the Great of Russia formalized in the Treaty of Georgievsk (1783), Georgia was ultimately unable to maintain its independence and was annexed by Russia in 1801.

After the Mongol and Timurid devastations, Georgia had fragmented into three major kingdoms—Kartli, Kakheti, and Imereti—and several semi-independent principalities including Samegrelo, Guria, and Svaneti. Each of these polities was too weak to resist the encroaching empires alone, and their periodic attempts at unification were undermined by internal rivalries and foreign interference. The three kingdoms developed distinct political cultures: Kartli was the ancient heartland with its capital at Tbilisi, Kakheti was the rich agricultural region to the east, and Imereti was the western kingdom centered on Kutaisi.

King Rostom (r. 1633–1658) of Kartli exemplifies the pragmatic accommodation that many Georgian rulers were forced to adopt during this period. A Muslim convert who served as a Persian-appointed governor, Rostom nonetheless worked to preserve Georgian cultural and religious institutions within the constraints of Persian overlordship. His reign was marked by relative peace and stability, achieved through careful diplomatic maneuvering between Persian and Ottoman interests. Rostom's story is a reminder that survival sometimes required painful compromises, and that not all Georgian rulers could afford the luxury of heroic defiance.

Vakhtang VI the Lawgiver (1675–1737) was one of the most remarkable figures of this era. A scholar-king who compiled Georgia's first systematic legal code (the "Dasturlamali" or "Book of Laws"), Vakhtang VI also reformed the tax system, sponsored the first printed books in Georgian, and maintained a court that attracted scholars and artists. His legal code standardized judicial practices across the kingdom and remained influential for generations. Despite his achievements, Vakhtang's reign ended tragically: after attempting to align with Russia against Persia, he was betrayed by the Russian government, which failed to provide the promised military support. He spent his final years in exile in Russia, dying in 1737 without ever returning to Georgia. Vakhtang VI is an excellent secondary hero for this era, whose "Legal Code" ability could provide systematic bonuses to tower efficiency and economy generation.

King Erekle II (Heraclius II, r. 1744–1798 in Kakheti, 1762–1798 in Kartli-Kakheti) stands as the towering figure of this period. He united the eastern Georgian kingdoms of Kartli and Kakheti into a single state, attempted to modernize the military and economy, and navigated the impossible diplomatic situation between rival empires. His alliance with Catherine the Great of Russia, formalized in the Treaty of Georgievsk on July 24, 1783, was supposed to guarantee Russian military protection for Georgia in exchange for Georgia surrendering its foreign policy autonomy. The treaty proved disastrous: when Agha Mohammad Khan of the Qajar Persian dynasty invaded Georgia in 1795, Catherine the Great withdrew Russian forces, leaving Erekle to face the Persian army alone.

The Battle of Krtsanisi (1795) is one of the most tragic events in Georgian history. Agha Mohammad Khan, infuriated by Erekle's alliance with Russia, marched on Tbilisi with a large army. Despite Erekle's desperate defense and the heroic resistance of the Georgian garrison, the city was overrun and completely destroyed. The Persians massacred the population, burned the city to the ground, and carried off thousands of captives. Tbilisi was reduced to smoldering ruins, and Erekle's kingdom never recovered. The destruction of Tbilisi directly led to Georgia's incorporation into the Russian Empire: with no other options remaining, the Georgian nobility petitioned for Russian annexation, which was formalized in 1801 after Erekle's death.

A crucial precursor to the Georgievsk betrayal was the earlier betrayal by Russian General Totleben at the Battle of Aspindza in 1770. During a joint Russo-Georgian military campaign against the Ottoman fortress of Akhaltsikhe, Totleben secretly withdrew his Russian forces, abandoning Erekle on the battlefield. Despite this betrayal, the Georgians won the Battle of Aspindza, but the incident foreshadowed the broader pattern of Russian unreliability as an ally. The Totleben betrayal should be referenced in the game's educational content as essential context for understanding Erekle's tragic alliance with Russia.

> **v4.0 Addition:** The v3.0 Bible omitted several significant figures and events from this era, including Vakhtang VI the Lawgiver, the Totleben betrayal at Aspindza (1770), and the nuanced reality of Erekle II's diplomatic efforts (which were ultimately tragic rather than skillful). These have been added.

Towers in this era reflect the evolving military technology of the early modern period. Gunpowder weapons appear for the first time, alongside traditional Georgian defensive structures. Maps are set in the three major Georgian kingdoms of the period: Kartli, Kakheti, and Imereti, each with its own architectural style and strategic challenges. Enemy waves alternate between Ottoman and Persian forces, each with distinct unit compositions and attack patterns that require different defensive strategies.

King Erekle II is the primary hero of this era, providing abilities that combine military leadership with the desperate nature of his diplomatic situation. His "Desperate Alliance" ability temporarily boosts all tower damage but at the cost of reduced tower HP, reflecting the double-edged nature of his Russian alliance. His "Last Stand" ultimate ability provides a massive but brief power surge to all defenses, representing the heroic but ultimately futile defense of Tbilisi at Krtsanisi. Vakhtang VI serves as a secondary hero with "Legal Code" abilities that boost tower efficiency and resource generation. The educational content covers the Treaty of Georgievsk and its betrayal, the Totleben betrayal at Aspindza, the Battle of Krtsanisi, Vakhtang VI's legal reforms, and the broader context of Georgian kingdoms caught between empires.

3.7 Era 6: Russian Empire (1801 – 1918 AD)

The Russian Empire era covers the period from Georgia's annexation by Russia in 1801 to the collapse of the empire in 1918. The annexation was formalized by Tsar Paul I in 1801, who violated the Treaty of Georgievsk by unilaterally incorporating the Kingdom of Kartli-Kakheti into the Russian Empire after the death of King George XII. What had been promised as a protective alliance became outright colonial rule. Despite the loss of political independence, this era was marked by a remarkable cultural renaissance as Georgian national identity was consciously preserved, strengthened, and modernized through literature, education, and political organization.

The early decades of Russian rule were characterized by the dismantling of Georgian institutions. The Georgian Orthodox Church lost its autocephaly (self-governance) in 1811 and was subordinated to the Russian Exarchate, a deeply resented measure that struck at the heart of Georgian cultural autonomy. The Georgian nobility was integrated into the Russian Imperial aristocracy, with many Georgian princes serving as officers in the Russian army and administrators in the Imperial bureaucracy. While this provided career opportunities for individual Georgians, it also meant that the natural leadership of the Georgian nation was co-opted by the colonial power.

The most prominent Georgian military figure of this era was Prince Pyotr Bagration (1765–1812), a Georgian-born Russian general who became one of the most celebrated commanders of the Napoleonic Wars. Bagration was descended from the Bagrationi royal dynasty and entered Russian military service as a teenager. He distinguished himself in numerous campaigns against the French, including Austerlitz (1805), Eylau (1807), and Friedland (1807), earning a reputation for aggressive tactics and fearless leadership. His finest hour came at the Battle of Borodino (1812), where he commanded the Russian left wing against the main French attack. Bagration held his position against repeated massive assaults but was mortally wounded during the battle. His death made

him a national hero in both Georgia and Russia, and his tactical mastery of defensive warfare makes him an ideal hero for this era.

> **v4.0 Correction:** The v3.0 Bible listed "Gia Abesalomashvili" as a potential military hero for this era. Extensive research found zero historical references to any figure by this name; it appears to be fabricated. This has been replaced with Prince Pyotr Bagration, a genuine and historically significant Georgian-born military commander from the Russian Empire era.

The Caucasian War (1817–1864), particularly the resistance of Imam Shamil in the mountainous regions of the North Caucasus, forms a significant military context for this era. While the war was primarily fought in Dagestan and Chechnya rather than in the Georgian heartland, it had major implications for Georgia: Russian military installations were built on Georgian territory, Georgian troops were conscripted into the Russian army, and the eventual Russian victory opened the Caucasus Mountains to Russian colonization. The war also brought the last major military resistance by indigenous Caucasian peoples against Russian expansion, a theme with deep resonance for Georgian players.

The second half of the 19th century saw the Georgian national awakening, a cultural and political movement aimed at preserving Georgian identity and achieving greater autonomy within the Russian Empire. The leading figure of this movement was Ilia Chavchavadze (1837–1907), a writer, journalist, lawyer, and public figure who is universally regarded as the father of modern Georgian nationalism. Chavchavadze founded the Tergdaleulebi (those who drank from the Terek) circle of Georgian intellectuals, established Georgia's first national bank to provide credit to Georgian peasants, campaigned for the preservation of the Georgian language in education, and used his literary works to inspire national consciousness. He was assassinated on August 30, 1907, near Tsitsamuri by assailants widely believed to be connected to Bolshevik revolutionaries. His murder remains one of the most controversial events in Georgian history and is seen by many Georgians as an early indication of the Bolshevik threat to Georgian democracy.

The era also saw major infrastructure development, including the construction of the Transcaucasian Railway, which connected the Black Sea port of Poti to the Caspian Sea port of Baku via Tbilisi. The railway was built in stages: the Poti–Tbilisi section opened in 1872, the extension to Baku was completed in 1883, and further connections were added over subsequent decades. The railway transformed Georgia's economy, making Tbilisi a major trading hub between Europe and Asia. However, the economic benefits were uneven, with many Georgian peasants remaining in poverty while Russian and Armenian merchants dominated the new commercial opportunities.

The 1905 Russian Revolution brought political upheaval to Georgia, with strikes, protests, and armed confrontations between revolutionaries and Tsarist forces. Georgia was one of the centers of revolutionary activity in the Russian Empire, and the Menshevik faction of the Russian Social Democratic Labor Party was particularly strong in Georgia. The revolution was ultimately suppressed, but it laid the groundwork for the democratic movement that would produce the First Democratic Republic after the collapse of the Tsarist regime in 1917.

Gameplay in this era introduces industrial-age military technology. Towers include fortified machine gun positions, artillery emplacements, and barricaded urban streets. Enemy waves represent the various conflicts of the period: the Caucasian War against Imam Shamil's forces, the 1905 Russian Revolution, and the chaos of World War I on the Caucasus Front, where Georgian troops fought in the Russian Imperial Army against the Ottoman Empire. Maps are set in rapidly modernizing Tbilisi, the mountain villages of Svaneti, the Black Sea port city of Batumi, and the military fortresses along the Caucasus frontier.

The heroes of this era are a combination of military and cultural figures. Prince Pyotr Bagration provides defensive military abilities, with his "Borodino Defense" ultimate ability creating an impenetrable shield around all towers for a brief duration, reflecting his legendary defensive stand against the French army. Ilia Chavchavadze provides abilities that boost tower efficiency and resource generation, reflecting his role as a national leader and modernizer. His "National Awakening" ultimate ability temporarily maximizes the output of all economy-generating structures on the map. The educational content covers the annexation of Georgia, the loss of church autocephaly, Bagration's military career, the Caucasian War, the Georgian national awakening, the Transcaucasian Railway, the 1905 Revolution, and Chavchavadze's assassination and legacy.

3.8 Era 7: First Democratic Republic (1918 – 1921 AD)

The First Democratic Republic of Georgia (1918–1921) was the first modern democratic state in the Caucasus and one of the first democratic republics to emerge from the ruins of the Russian Empire. Led by the Social Democratic (Menshevik) government of Noe Jordania (also transliterated as Zhordania), Georgia declared independence on May 26, 1918, and spent three chaotic years building a multi-ethnic, multi-party democratic state while fighting wars on multiple fronts and struggling against internal and external threats.

The path to independence was turbulent. Following the February Revolution of 1917 that overthrew the Tsar, the Transcaucasian Democratic Federative Republic was established in April 1918 as a short-lived federation of Georgia, Armenia, and Azerbaijan. This federation dissolved after just one month due to

irreconcilable differences between the three national groups, and Georgia declared its own independence on May 26, 1918. The brief existence of the Transcaucasian Federation is historically significant as an early attempt at regional cooperation in the South Caucasus, and its failure foreshadowed the ethnic and political tensions that continue to shape the region.

Despite its brief existence, the First Republic achieved remarkable accomplishments. The constitution adopted in 1921 guaranteed universal suffrage (making Georgia one of the first countries in Europe to grant women the right to vote), minority rights, judicial independence, land reform, and workers' protections. The republic's democratic institutions, including a freely elected parliament and a free press, functioned even under the extreme pressure of war and economic crisis. Georgia's democratic experiment was remarkable not only for its progressive policies but also for its context: it was built in a region with no democratic tradition, surrounded by hostile forces, during a period of global upheaval.

International recognition was partial. Georgia received de jure recognition from several major powers, including Britain and France, though its application for League of Nations membership was postponed and never approved. The League of Nations "fully supported Georgia's independence" but did not admit it as a member, a distinction that the game should accurately represent. British military forces were present in Georgia from 1918 to 1920 as part of the Allied intervention in the Russian Civil War, providing a measure of protection against Bolshevik expansion while also pursuing British strategic interests in the Caucasus. The British withdrawal in 1920 removed the last obstacle to a Bolshevik invasion.

> **v4.0 Correction:** The v3.0 Bible stated Georgia achieved "recognition by major international powers including the League of Nations." This wording could mislead players into thinking Georgia was a League member. In reality, Georgia's application for membership was postponed and never approved. Additionally, the Transcaucasian Federation and British involvement (1918–1920) were omitted from v3.0 and have been added.

The republic fought several defensive wars simultaneously: the Armeno-Georgian War of December 1918 over disputed border territories (a bitter conflict between two newly independent nations that should have been natural allies), the White Russian threat from General Denikin's Volunteer Army (Georgian forces under General Giorgi Mazniashvili successfully defended Georgian territory against the Whites), and border conflicts with Azerbaijan. These overlapping conflicts stretched the republic's limited military resources to the breaking point.

The era ended when the Bolshevik Red Army invaded on the night of February 11–12, 1921, at Ordzhonikidze's instigation. The Red Army's overwhelming

numerical and material superiority made the outcome inevitable, though Georgian forces fought bravely in the three-week campaign. The democratic government went into exile, first to Istanbul and then to France, where the Menshevik government-in-exile continued to advocate for Georgian independence throughout the Soviet period. The Bolshevik occupation was formalized on February 25, 1921, a date now commemorated in modern Georgia as Soviet Occupation Day.

Gameplay in this era features early 20th-century military technology: rifles, machine guns, and artillery. Enemy waves represent the multiple threats faced by the republic simultaneously, creating a unique gameplay challenge where players must defend against attacks from different directions with different enemy types. Maps are set in Tbilisi, the Black Sea coast, and the border regions where the republic fought its defensive wars. The hero is Noe Jordania, whose abilities focus on diplomatic coordination that temporarily strengthens all towers and generates emergency resources. His "Democratic Resilience" ultimate ability temporarily makes all towers on the map significantly more powerful, representing the strength that Georgia's democratic institutions provided even under extreme duress. Educational pop-ups cover the constitution, the Transcaucasian Federation, the international recognition campaign, the British military presence, the multiple border wars, and the tragic circumstances of the Bolshevik invasion.

3.9 Era 8: The Soviet Century (1921 – ~1991 AD)

The Soviet Century covers the seventy years of Georgian history within the Soviet Union. This era is presented with historical honesty, neither glorifying the Soviet period nor erasing its complexities. Georgia experienced both severe repression and significant modernization during this time, and the game must represent both aspects accurately to fulfill its educational mission.

The most controversial figure of this era is Joseph Stalin (born Ioseb Jughashvili in Gori, Georgia, 1878–1953), who rose from Georgian origins to become the Soviet dictator responsible for millions of deaths. Stalin's relationship with Georgia was complex: he showed little personal loyalty to his homeland, ordering the brutal suppression of the 1924 Georgian uprising against Soviet rule and overseeing purges that devastated the Georgian Communist Party and intelligentsia. Yet many Georgians felt a complicated pride in his rise to power, and his death in 1953 was mourned in Tbilisi. The game handles Stalin by presenting historical facts without commentary, allowing players to engage with this difficult history through the gameplay context. Stalin is not a playable hero character but is referenced in historical context and educational content.

The Great Purge of 1937–1938 disproportionately affected Georgia. A significant proportion of the Georgian Communist Party, intelligentsia, and military officers were executed or sent to the Gulag. The purges were particularly devastating because they eliminated an entire generation of Georgian leaders, including many who had been loyal Communists. The educational content for this era should note that the purges were not merely a Moscow-imposed phenomenon but were also carried out by Georgian Bolsheviks against their fellow Georgians, adding a layer of tragedy and moral complexity to the story.

World War II had a massive impact on Georgia. Approximately 700,000 Georgians served in the Soviet armed forces during the war, of whom approximately 190,000 were killed in action. Georgian soldiers fought in every major theater of the war, from the defense of Moscow and Stalingrad to the liberation of Europe. Several Georgian formations distinguished themselves in combat, and individual Georgian soldiers received numerous awards for valor. The war also saw the entire Chechen and Ingush populations deported from the neighboring North Caucasus in 1944, an event that had lasting consequences for the region's ethnic politics.

> **v4.0 Correction:** The v3.0 Bible stated "approximately 350,000 Georgian soldiers died in WWII." Authoritative sources confirm that approximately 700,000 Georgians served, of whom approximately 190,000 were killed in action. The figure has been corrected. Some sources cite higher figures for total Georgian losses including non-combat deaths, but 190,000 is the widely accepted figure for military deaths in action.

The 1956 Tbilisi Massacre (March 9, 1956) is one of the most significant and traumatic events of the Soviet period in Georgia. Following Nikita Khrushchev's secret speech denouncing Stalin's cult of personality, mass demonstrations erupted in Tbilisi, driven by a complex mixture of Georgian national pride, anger at the denigration of a fellow Georgian, and broader grievances against Soviet rule. The protests began as student demonstrations but grew to include thousands of people. Soviet authorities responded by sending tanks and troops into the city, who opened fire on the unarmed demonstrators. Estimates of the death toll range from dozens to over a hundred, with hundreds more wounded. The massacre was suppressed in Soviet media and was not publicly discussed in Georgia until the late 1980s. This event is essential for understanding the complex relationship between Georgians and the Soviet state, and between Georgians and Stalin's legacy.

> **v4.0 Addition:** The 1956 Tbilisi Massacre was entirely omitted from v3.0. This event is critical for the Soviet era and reveals the complex relationship between Georgians and Stalin's legacy. It has been added as a major historical reference in this era.

The 1978 language protests represent another pivotal moment in Soviet Georgian history. On April 14, 1978, mass protests erupted in Tbilisi when the Soviet authorities proposed a new draft constitution for the Georgian SSR that would have removed Georgian's status as the sole state language. Tens of thousands of Georgians, including students, intellectuals, and ordinary citizens, took to the streets in a remarkable display of popular resistance. By the end of the day, Moscow had acquiesced to popular pressure and the disputed clause was withdrawn. April 14 is now celebrated in Georgia as the Day of the Georgian Language. This event demonstrated that even under Soviet rule, Georgians were willing to mobilize to protect their cultural identity, and it foreshadowed the national independence movement of the late 1980s.

The late 1980s saw the growth of the Georgian national independence movement, culminating in the declaration of sovereignty in 1989 and independence in 1991. The movement was fueled by a combination of national consciousness, economic grievances, and the broader crisis of the Soviet state. Key events included the 1989 Tbilisi tragedy (Soviet troops attacked peaceful demonstrators with spades and toxic gas, killing 19 people, mostly young women), which galvanized the independence movement and delegitimized Soviet authority in Georgia.

Towers in this era reflect Soviet military technology: reinforced concrete bunkers, anti-aircraft positions, and fortified checkpoints. Enemy waves represent the various conflicts that touched Georgia during the Soviet period, including World War II (where Georgian soldiers fought on the Eastern Front), anti-Soviet partisan resistance in the mountains, and the environmental and social disruptions of late Soviet industrialization. Maps are set in Soviet-era Tbilisi, industrial zones, and mountain villages that served as centers of resistance.

The hero selection for this era requires careful handling. Historical figures from the Soviet period are presented factually, with educational pop-ups providing context about their roles and the human cost of their actions. Stalin is not a playable hero character but is referenced in educational content. Potential heroes include Merab Kostava (a dissident and national liberation figure who was imprisoned multiple times for his activism) and World War II Georgian military heroes. The educational content covers the Great Purge, World War II, the 1956 Tbilisi Massacre, the 1978 language protests, the 1989 Tbilisi tragedy, and the growing independence movement of the late 1980s.

3.10 Era 9: Modern Georgia (~1991 – Present)

The final era brings players to the modern Republic of Georgia, from the chaotic first years of independence through the Rose Revolution of 2003 to the present

day. This era covers the 1991–1993 civil war and war in Abkhazia, the economic collapse of the 1990s, the peaceful Rose Revolution that ousted the Shevardnadze government, the 2008 war with Russia, and the ongoing process of democratic consolidation and Euro-Atlantic integration. It is the most sensitive era in the game and requires the most careful handling of contemporary political issues.

Georgia declared independence from the dissolving Soviet Union on April 9, 1991, under the leadership of Zviad Gamsakhurdia, Georgia's first popularly elected president. Gamsakhurdia's presidency was brief and turbulent; his increasingly authoritarian rule and nationalist rhetoric alienated ethnic minorities and moderate Georgians alike, leading to a violent coup in December 1991–January 1992. The subsequent civil war between Gamsakhurdia's supporters (Zviadists) and the new government devastated western Georgia and resulted in Gamsakhurdia's death under mysterious circumstances in 1993.

Former Soviet Foreign Minister Eduard Shevardnadze was invited to lead the country in 1992 and was elected president in 1995. His rule brought relative stability but was marred by corruption, economic stagnation, and the loss of Georgian control over Abkhazia and South Ossetia. The Abkhaz War (1992–1993) resulted in a devastating Georgian defeat and the ethnic cleansing of approximately 200,000–250,000 Georgians from Abkhazia, who remain displaced to this day. The war in South Ossetia (1991–1992) ended with a Russian-brokered ceasefire that left the region effectively under Russian control. These territorial losses remain open wounds in Georgian national consciousness and are central to the country's current political orientation toward the West.

The Rose Revolution of November 2003 was a watershed moment. Mass protests against parliamentary election fraud, led by Mikheil Saakashvili and other opposition leaders, culminated in the peaceful resignation of Shevardnadze. Saakashvili was elected president in January 2004 and launched an ambitious reform program: fighting corruption, modernizing the police, restructuring the economy, and pursuing NATO and EU membership. However, his government became increasingly authoritarian over time, using riot police against opposition demonstrations and concentrating power in the presidency. Saakashvili's legacy remains deeply divisive in Georgia: credited with modernization and anti-corruption reforms, but criticized for authoritarian tendencies and the catastrophic 2008 war with Russia.

The Russo-Georgian War of August 2008 was a defining moment for modern Georgia. Following months of escalating tensions and provocations in South Ossetia, full-scale war broke out on the night of August 7–8, 2008. Russian forces quickly overwhelmed the Georgian military, advancing beyond South Ossetia into Georgian territory proper, occupying the cities of Gori, Poti, and Zugdidi. The war lasted five days and ended with a Russian-announced ceasefire. Russia

subsequently recognized the independence of Abkhazia and South Ossetia, a move recognized only by a handful of countries. The war demonstrated the limits of Western support for Georgia and the reality of Russian military power in the region. It is important to note that the primary adversary in this conflict was the Russian military, not merely "separatist forces," though the game should acknowledge the role of South Ossetian and Abkhaz militias as secondary enemies.

The post-2008 period has seen Georgia's continued pursuit of Euro-Atlantic integration. The country signed an Association Agreement with the European Union in 2014, achieved visa-free travel to the Schengen Area in 2017, and was granted EU candidate status in December 2023. However, Georgia's democratic trajectory has been complicated by the growing authoritarian tendencies of the Georgian Dream government, which has been in power since 2012. The passage of a "foreign agent" law in 2024, modeled on Russia's similar legislation, sparked massive protests and led the European Union to freeze Georgia's candidate status in July 2024. This ongoing political crisis represents the most current chapter in Georgia's struggle to define its identity as a Western-oriented democracy in a region of renewed Russian assertiveness.

Towers in this era represent modern military technology: NATO-standard defensive systems, drone surveillance platforms, and fortified military outposts. The tower descriptions should accurately reflect Georgia's actual military capabilities, which are based on Western-supplied equipment rather than indigenous advanced systems. Enemy waves reflect the conflicts of modern Georgia: Russian conventional military forces, separatist militias, and irregular fighters. Maps are set in modern Tbilisi, the conflict zones of South Ossetia and Abkhazia, and the strategic Georgian Military Highway connecting Tbilisi to Russia through the Caucasus Mountains.

The heroes of this era are drawn from modern Georgian history. Their identities should be handled with sensitivity to ongoing political controversies. The educational content covers the challenges of post-Soviet state-building, the Abkhaz and South Ossetian conflicts, the achievements and controversies of the Saakashvili era, the 2008 Russo-Georgian War, Georgia's Euro-Atlantic integration, and the 2023–2024 political crisis over the foreign agent law and frozen EU candidacy. The final levels of this era serve as the game's climax, bringing together all the mechanical systems and historical themes developed across the previous nine eras.

4. Tower System

The tower system is the mechanical core of Sakartvelo Defenders. Ten distinct tower types provide the player's defensive toolkit, each filling a specific strategic role. Towers are introduced gradually across the ten eras, with each era introducing one new tower type that reflects the military technology of that historical period. Every tower receives era-specific visual skins to maintain historical authenticity while preserving the underlying game mechanics that players have mastered.

The tower progression follows a clear design principle: newer towers add strategic options rather than rendering older ones obsolete. An Archer Tower from Era 0 remains viable in Era 9 because it occupies a unique niche (fast single-target damage at low cost) that no later tower perfectly replicates. This ensures that all ten tower types remain relevant throughout the full 200-level campaign, rewarding players who have mastered diverse tower compositions.

Tower Type	Role	Introduced	Era Skin Example
Archer Tower	Basic single-target DPS	Era 0	Colchian Watchtower / Iberian Fort
Catapult/ Trebuchet	Area-of-effect splash damage	Era 0	Stone Thrower / Siege Engine
Wall/Barrier	Path blocking, enemy redirecting	Era 0	Wooden Palisade / Stone Wall
Cavalry Outpost	Fast patrol, intercepting enemies	Era 1	Iberian Horsemen / Georgian Knights
Siege Tower	Heavy damage, slow attack, anti-structure	Era 2	Fortified Gatehouse / Battlement
Church Shrine Tower	Buff/debuff, crowd control	Era 3	Monastery Shrine / Cathedral
Gunpowder Tower	Long-range, piercing projectiles	Era 5	Cannon Emplacement / Musketry
Industrial Tower	Rapid-fire, multi-target suppression	Era 6	Machine Gun Nest / Artillery
Bunker Tower	Heavy armor, high survivability	Era 8	Soviet Bunker / Concrete Pillbox
Tech Tower	Precision strikes, drone support	Era 9	Modern Outpost / Drone Platform

The Church Shrine Tower (introduced in Era 3) deserves special historical explanation. During the Age of Invasions and the Arab occupation, the Georgian Orthodox Church was the primary institution that preserved Georgian national identity, language, and culture. Churches and monasteries served as both spiritual centers and physical fortresses, with many monasteries built as fortified complexes that could withstand military attack. The Church Shrine Tower's abilities reflect this dual spiritual and military role: "Bell Toll" slows all enemies in a wide radius (representing the morale effect of church bells rallying defenders), "Sanctuary" creates a temporary shield around nearby towers (representing the physical protection of fortified church walls), and "Divine Light" at Level 3 increases the damage of all nearby towers (representing the spiritual inspiration that fueled Georgian resistance during centuries of occupation). The educational content should explain that while the Church Shrine Tower's "magic" effects are game-ified, they represent the very real role of the Georgian Orthodox Church in sustaining national resistance and cultural survival.

4.1 Tower Upgrade System

Each tower can be upgraded through three levels, with each upgrade increasing damage, range, attack speed, or adding special effects. Upgrade costs scale exponentially, encouraging players to invest in a focused defense rather than spreading resources thinly across many weak towers. At maximum upgrade level (Level 3), towers gain a unique special ability specific to their type: Archer Towers gain a chance for critical hits, Catapults gain increased splash radius, Walls gain damage reflection, Cavalry Outposts gain a charge attack, Siege Towers gain armor penetration, Church Shrine Towers gain "Divine Light" (damage boost aura), Gunpowder Towers gain explosive rounds, Industrial Towers gain area suppression, Bunker Towers gain self-repair, and Tech Towers gain target-locking precision strikes.

Tower upgrades are purchased with in-game resources earned from defeating enemies. This resource economy is entirely separate from the SAKART token economy and functions as the core gameplay loop driver. Players must make strategic decisions about when to invest in upgrades versus placing additional towers, creating meaningful choices every level. The upgrade system is designed to feel impactful: a Level 3 Archer Tower should feel dramatically more powerful than a Level 1, both visually and mechanically.

4.2 Tower Synergy System

Synergies are activated when specific tower combinations are placed within a 3-tile adjacency radius of each other. Synergies provide powerful bonuses that

encourage diverse tower compositions and strategic map planning. Each synergy combination is introduced in the era where its component towers are most thematically appropriate, with new combinations becoming available as players unlock more tower types.

The synergy system creates emergent strategic depth. A player who places an Archer Tower near a Cavalry Outpost might discover a synergy that increases both units' attack speed, creating a rapid-response kill zone. Another player might combine a Church Shrine Tower with a Wall/Barrier to create a "Holy Fortress" synergy that dramatically increases wall durability and adds a slowing aura, reflecting the historical role of fortified monasteries. The system rewards experimentation and map-specific planning, ensuring that no single strategy dominates across all levels.

With ten tower types, there are 45 possible two-tower pairs and 120 possible three-tower triples, though only a curated subset of the most strategically interesting and thematically appropriate combinations will be implemented. Each era introduces approximately 2–3 new synergy combinations, building up to a rich strategic toolkit by the final era. Named synergies should reflect Georgian historical and cultural themes where possible, such as "Didgori Formation" (a military synergy reflecting David IV's tactics), "Tamar's Grace" (a shrine-based defensive synergy), or "Caucasus Wall" (a barrier-heavy fortification synergy).

5. Enemies

Enemy design in Sakartvelo Defenders follows two core principles: historical authenticity and mechanical clarity. Every enemy type is based on a real historical military unit or cultural archetype from Georgian history, modified as needed for gameplay balance. Enemies are visually designed to be instantly distinguishable, with clear silhouettes, color coding by threat level, and animation that communicates their behavior (fast enemies move erratically, armored enemies have deliberate, heavy animations).

Enemy HP and damage scale progressively across the 200 levels, but the scaling is carefully calibrated to maintain challenge without requiring numerical inflation that makes earlier towers irrelevant. Instead of simply increasing HP values, later eras introduce enemy types with new mechanics: shields that block frontal damage, healing abilities that restore HP to nearby enemies, flying units that bypass ground-based tower ranges, and boss enemies with unique phase-based attack patterns.

5.1 Enemy Categories

Enemies are divided into five broad categories that define their mechanical behavior:

- **Infantry** units follow standard path movement at moderate speed and HP, forming the backbone of most waves. Era-specific examples include Colchian tribal warriors (Era 0), Arab foot soldiers (Era 2), Mongol infantry (Era 4), Ottoman janissaries (Era 5), and Soviet conscripts (Era 8).
- **Cavalry** units move faster and have higher HP but follow the same paths as infantry. Era-specific examples include Iberian mounted archers (Era 1), Seljuk Turk horsemen (Era 2), Mongol horse archers with hit-and-run tactics (Era 4), Persian qizilbash cavalry (Era 5), and Cossack cavalry (Era 6).
- **Siege** units are slow but devastating, capable of damaging player towers directly. Era-specific examples include Greek siege engines (Era 0), Arab siege towers (Era 2), Mongol catapults (Era 4), Ottoman bombards (Era 5), and Russian artillery (Era 6).
- **Flying** units ignore path constraints and can only be targeted by towers with anti-air capabilities. These include mythological creatures in early eras (Colchian eagles, devi that fly) and modern aircraft and drones in later eras.
- **Boss** enemies appear at the end of specific levels as challenging multi-phase encounters (see Section 5.2).

5.2 Boss Encounters

Boss encounters appear at the end of every 5th level within each era (levels 5, 10, 15, and 20). Each boss is a named historical or mythological figure with a multi-phase fight that requires specific strategies to defeat. Bosses have dramatically higher HP than standard enemies and feature unique abilities that challenge the player's tower placement and resource management.

Era	Boss	Type	Special Ability
0	Colchian Dragon	Mythological	Regenerating heads (hydra mechanic); requires coordinated multi-tower attacks
0	Devi Chief	Mythological	Three-headed ogre; each head has a different attack; heads regenerate unless all destroyed simultaneously
1		Historical	

	Sassanid Persian General		Summons reinforcing infantry waves; shield that blocks frontal damage
1	Roman Centurion Commander	Historical	Formation ability that buffs nearby infantry; Roman testudo shield reduces all incoming damage
2	Emir Isma'il of Tbilisi	Historical	Summons reinforcing waves; desert heat aura that slowly damages nearby towers
2	Seljuk Warlord	Historical	Hit-and-run attacks that damage towers then retreat; cavalry summon ability
3	Seljuk Il-Deniz	Historical	Massive army summon; coordinates multi-directional attacks
3	Khwarezmian Raider	Historical	Rapid movement; splits into smaller units when damaged
4	General Chormaqan	Historical	Speeds up all enemy movement across the map; Mongol horn ability that summons cavalry reinforcements
4	Timur (Tamerlane)	Historical	Area damage to nearby towers; devastation ability that permanently reduces tower damage in an area
5	Agha Mohammad Khan	Historical	Area damage to towers; firespread ability that damages multiple towers; "Scorched Earth" ultimate that destroys the lowest-HP tower on the map
5	Ottoman Grand Vizier	Historical	Janissary summon; bombardment ability that targets player's most valuable tower
6	Imam Shamil	Historical	Mountain ambush ability; guerrilla tactics that make enemies temporarily invisible
6	Tsarist General	Historical	Artillery barrage; reinforcement waves from multiple directions
7	Bolshevik Commissar	Historical	Propaganda ability that converts one tower to enemy side temporarily; Red Guard summon
8	Stalin (historical context)	Historical	Purge ability that randomly disables one tower for a duration; NKVD summon
8	Nazi Panzer Commander	Historical	Heavy armor; requires armor-piercing towers to damage effectively
9	Russian Tank Column	Historical	Extremely heavy armor; requires concentrated fire from multiple towers

9	Separatist Warlord	Historical	Guerrilla tactics; ambush spawns enemies behind tower lines

> **v4.0 Correction:** The v3.0 Bible listed "Subutai's proxy" as the Era 4 Mongol boss. Subutai himself passed through Georgia in 1220 on a reconnaissance mission, not as a conqueror. The actual Mongol commander who subjugated Georgia was General Chormaqan, who led the 1236–1243 conquest. The boss has been changed to Chormaqan, with Timur added as a second boss for the era's later levels.

6. Heroes

Heroes are special units that players deploy alongside their tower defenses. Each hero has a unique set of active abilities that the player triggers manually during combat, adding a layer of real-time decision-making to the tower defense formula. Heroes are era-specific, meaning they can only be used in levels belonging to their historical period. This design encourages players to engage with each era's historical context and ensures that hero abilities feel thematically appropriate.

The design philosophy is to provide at least one hero per era, with major eras featuring two to three heroes to reflect the abundance of notable historical figures. Both David IV and Queen Tamar are playable heroes in Era 3, representing the two most important figures of the Georgian Golden Age. Stalin is handled historically in Era 8, presented factually in educational content but not as a playable hero character.

Era	Hero	Type	Key Ability
0	Medea	Support/ Control	Poison damage + tower healing; "Colchian Alchemy" ultimate transforms nearby towers into enhanced versions
1	King Pharnavaz	Kingdom Buff	Economy boost, reduced tower costs; "Unification" passive reduces multi-sector placement costs
1	Saint Nino	Healer	Tower shield, enemy slow aura; "Vine Cross" ultimate heals all towers and creates a slowing zone
1	Vakhtang Gorgasali	Offense	Area damage boost for nearby towers; "Wolf's Head" charge damages enemies in a line
2		Consolidation	

	Bagrat III the Unifier		Tower stat boost; "Unification" reduces multi-sector placement costs; "Kingdom United" ultimate temporarily boosts all towers
3	David IV the Builder	Offense/ Leadership	Global damage buff, upgrade cost reduction; "Dzleva Sakvirveli" (Wonderful Victory) ultimate temporarily doubles all tower damage
3	Queen Tamar	Defense/ Economy	Resource generation, temporary tower invulnerability; "King of Kings" ultimate makes all towers invulnerable with maximum damage
3	Shota Rustaveli	Cultural Inspiration	Passive attack speed/damage boost for all towers; "Knight's Valor" ultimate temporarily maximizes all tower attack speed
3	Zakare Zakarian	Offense/ Command	Cavalry tower boost, coordinated strikes; "Caucasian Conquest" ultimate sends a massive cavalry wave against enemies
4	Queen Rusudan	Diplomacy/ Desperation	Emergency resource generation, enemy damage reduction; "Last Appeal" ultimate generates massive resources but reduces tower HP
4	George V the Brilliant	Restoration	Tower repair, stat restoration; "Reunification" ultimate repairs all towers and temporarily boosts stats to Golden Age levels
5	King Erekle II	Defense/ Desperation	"Desperate Alliance" boosts tower damage but reduces tower HP; "Last Stand" ultimate provides massive but brief power surge
5	Vakhtang VI the Lawgiver	Economy/Law	Tower efficiency boost, resource generation; "Legal Code" passive systematically improves all economy generation
6	Prince Pyotr Bagration	Defense/ Military	"Borodino Defense" ultimate creates impenetrable shield around all towers briefly; passive damage reduction for nearby towers
6	Ilia Chavchavadze	Economy/ Culture	Tower efficiency, resource generation; "National Awakening" ultimate

			maximizes all economy-generating structures
7	Noe Jordania	Diplomacy	Temporary global tower strength boost; "Democratic Resilience" ultimate temporarily makes all towers significantly more powerful
8	Merab Kostava	Resistance	Tower resilience boost, enemy debuff; "National Will" ultimate creates a powerful defensive aura around all towers
8	WWII Georgian Commander	Offense	Targeted strike ability; "Hero City" ultimate fortifies all towers in an area to maximum level temporarily
9	Modern Era Figure	Tech/Precision	Drone recon reveals hidden enemy paths; precision targeting ability

v4.0 Corrections: (1) "Gia Abesalomashvili" has been removed and replaced with Prince Pyotr Bagration, a genuine Georgian-born military hero. (2) "Zakaria Mkhargrdzeli" renamed to "Zakare Zakarian" using the standard academic name. (3) Shota Rustaveli's ability changed from "Morale Boost" to "Cultural Inspiration" to better reflect his historical role as a poet rather than a military commander. (4) Erekle II's ability changed from generic "diplomacy" to "Desperate Alliance/Last Stand" to better reflect the tragic outcome of his diplomatic efforts. (5) Bagrat III the Unifier, Queen Rusudan, George V the Brilliant, Vakhtang VI, and Merab Kostava added as new heroes.

6.1 Hero Abilities and Cooldowns

Each hero has three abilities: one basic ability on a short cooldown (15–20 seconds), one advanced ability on a medium cooldown (45–60 seconds), and one ultimate ability on a long cooldown (120–180 seconds). Basic abilities provide tactical benefits like minor damage buffs or healing. Advanced abilities can shift the flow of an individual wave. Ultimate abilities are game-changing effects that, when timed correctly, can save a failing defense or enable a perfect three-star run.

When a hero is defeated (all HP depleted), they enter a cooldown period before they can be redeployed. During this cooldown, the player must rely solely on their tower defenses. The cooldown duration scales with the hero's power level, ensuring that losing a powerful hero like Queen Tamar is more punishing than losing a support-oriented hero. This mechanic encourages players to position heroes strategically rather than exposing them to unnecessary damage.

6.2 Hero Progression

Heroes level up through gameplay by participating in battles. Each battle a hero participates in grants experience points based on the enemies defeated. Leveling up improves hero base stats (HP, damage) and reduces ability cooldowns. Hero levels are stored on-chain as part of the player's account data, ensuring that progression is persistent and portable. The maximum hero level is capped to prevent power inflation, with the cap set to ensure that a max-level hero is powerful but not capable of solo-carrying a level without tower support.

7. Maps

Each of the 200 levels is set on a unique map inspired by real Georgian landscapes, fortifications, and cities. Maps are designed with the isometric 2D perspective in mind, ensuring clear visibility of enemy paths, tower placement zones, and strategic elevation points. The art style for each map reflects the architectural and environmental characteristics of the era, creating a visual journey through Georgian history that parallels the mechanical progression.

Map design follows several key principles. First, every map must have at least three viable tower placement strategies to encourage replay and experimentation. Second, maps must feature at least one elevation mechanic, where towers placed on higher ground receive increased range, adding a spatial dimension to tower placement decisions. Third, each map must include at least one chokepoint where players can concentrate defensive fire, balanced by at least one wide-open section that requires spread defenses. These principles ensure strategic variety across all 200 levels.

7.1 Map Themes by Era

Era 0 maps feature prehistoric Colchian landscapes: dense forests along the Rioni River, gold-panning sites where sheep fleeces were used to trap gold particles, the ancient city of Vani with its temple complexes and defensive walls, and the Greek colonial ports of Phasis and Dioscurias on the Black Sea coast. The Colchian environment is lush, wet, and subtropical, with distinctive flora including Colchic boxwood forests and Rhododendron thickets that provide natural visual barriers and pathing options.

Era 1 maps showcase the early urban centers of Mtskheta (the ancient capital situated at the confluence of the Mtkvari and Aragvi rivers, one of the oldest continuously inhabited cities in the world), the cave city of Uplistsikhe (with its unique carved-stone layout providing pathing through tunnels and chambers),

the fortress of Armazi perched above Mtskheta, and the newly founded city of Tbilisi with its hot springs and growing urban landscape. The environment transitions from the subtropical west to the more arid eastern Georgian plains.

Era 2 maps depict fortified monasteries and mountain passes under siege from Arab and Seljuk forces. Key locations include the David Gareja monastery complex (carved into the semi-arid hillsides of southeastern Georgia, providing unique vertical map layouts), the Ananuri fortress on the Aragvi River, and the mountain passes of the Greater Caucasus where Georgian defenders held off invading armies. The Seljuk invasion maps of the late era reflect the devastation of the Didi Turkoba, with burnt villages and empty landscapes.

Era 3 maps feature the grand castles and cathedrals of the Golden Age, including Gelati Monastery and Academy (founded by David IV in 1106, a UNESCO World Heritage Site and center of learning), the cave city of Vardzia (constructed under George III and Tamar, a massive underground city carved into the mountain that could house 50,000 people), the fortress of Khertvisi, and the liberated city of Tbilisi. These maps are visually the most impressive in the game, reflecting the wealth and power of the Georgian Golden Age.

Era 4 maps show the devastation of the Mongol period: burning cities, abandoned fortresses, and desperate mountain retreats. A brief mid-era map set reflects George V's restoration, with rebuilt churches and restored fortifications, before Timur's invasions return the visual tone to destruction. The cave cities that served as refuges (Vardzia, Davit Gareja) feature prominently as defensive maps where players must protect civilian populations.

Era 5 maps are set in the three kingdoms of Kartli, Kakheti, and Imereti, with distinct architectural styles for each. Kartli maps feature Persian-influenced architecture, Kakheti maps showcase Georgian church architecture and vineyard landscapes, and Imereti maps feature more Byzantine-influenced buildings. The Battle of Krtsanisi map is one of the most emotionally impactful in the game, showing the burning of Tbilisi.

Era 6 maps feature rapidly modernizing Tbilisi with its mix of old and new architecture, the mountain villages of Svaneti with their iconic defensive towers, the Black Sea port city of Batumi, and military installations along the Caucasus frontier. The Transcaucasian Railway features as a map element, with trains carrying resources along the tracks.

Era 7 maps depict the war-torn landscapes of the independence period, with trenches, barricaded streets, and the provisional government buildings of Tbilisi. Maps reflect the multi-front nature of the DRG's defensive wars, with enemy approaches from different directions.

Era 8 maps showcase Soviet-era industrial zones, brutalist architecture, collective farms, and mountain villages that served as centers of resistance. The WWII maps feature Eastern Front battlefields where Georgian soldiers fought. Later maps show the growing independence movement, with protest sites in central Tbilisi.

Era 9 maps present modern Georgia with contemporary military installations and urban environments. Key locations include central Tbilisi, the conflict zones of South Ossetia and Abkhazia, the strategic Georgian Military Highway connecting Tbilisi to Russia through the Caucasus Mountains (one of the world's most spectacular and dangerous roads), and the Black Sea coast near Batumi.

7.2 Elevation Mechanic

The elevation mechanic is a core map feature that rewards strategic tower placement. Maps are divided into ground-level and elevated positions, with higher positions providing a range bonus to any tower placed there. The range bonus is approximately 15–20%, giving elevated towers a meaningful but not overwhelming advantage. Maps are designed so that the best elevated positions are also the most exposed or hardest to defend, creating risk-reward decisions for players.

Additionally, arrow-type and projectile towers benefit disproportionately from elevation, gaining both increased range and a damage bonus due to the improved angle of attack. This means an Archer Tower on a hill is dramatically more effective than one on flat ground, encouraging players to prioritize elevated positions for their projectile-based defenses while using melee-adjacent towers like Wall/Barrier units to control ground-level approaches. The elevation mechanic is particularly important on maps set in Georgia's mountainous terrain, where the country's dramatic topography provides natural defensive positions.

8. Tutorial

The tutorial is integrated into the first five levels of Era 0, teaching core mechanics organically through gameplay rather than separate tutorial screens. The philosophy is that players learn by doing, with contextual hints appearing only when the game detects that a player is struggling. The tutorial levels are designed to be completable by any player regardless of experience with tower defense games, while still being engaging enough to retain experienced players.

Level 1 introduces tower placement: the player is given a single Archer Tower and must place it to defend against a slow, weak enemy wave. Level 2 introduces the economy system, requiring players to earn resources from kills and invest

them in a second tower. Level 3 introduces tower upgrades by presenting a wave that cannot be defeated with Level 1 towers alone. Level 4 introduces the hero system with Medea's basic ability. Level 5 introduces scrolls as a consumable resource and features the first mini-boss encounter.

By the end of these five levels, players have mastered all core systems and are ready for the full campaign experience. Each tutorial level includes a brief historical pop-up between levels that introduces the world of Ancient Colchis. These pop-ups are concise (2–3 sentences) and focus on creating intrigue rather than delivering encyclopedic information. The deeper educational content is reserved for the main campaign, where players who are already invested in the gameplay will be more receptive to historical context.

9. Progression

Progression in Sakartvelo Defenders operates on two parallel tracks: gameplay progression and economic progression. Gameplay progression is the journey through 200 levels across 10 eras, with each level unlocked by completing the previous one. Economic progression is the accumulation of SAKART tokens earned from star completions, which can be used to purchase cosmetic items, mastery scrolls, and access to competitive features.

The progression system is designed to be generous with gameplay content while strictly controlling economic rewards. All 200 levels are accessible to every player without any payment requirements. The SAKART economy is driven entirely by in-game achievement: each star earned on a level grants a one-time token reward, with higher stars granting larger rewards. Because each star can only be earned once per level, there is zero incentive for replay farming, and the total token supply is mathematically bounded by the number of players and levels.

9.1 Era Unlock Requirements

Eras are unlocked sequentially. To unlock the next era, a player must earn at least 1 star (the basic completion star) on a minimum number of levels in the current era. The requirement is set at 15 out of 20 levels, allowing players to skip the most difficult levels while still demonstrating sufficient mastery. This ensures that players experience most of each era's content before moving forward, maintaining the chronological educational narrative without creating frustrating progression blockers.

The era unlock requirement is intentionally permissive rather than restrictive. The game's philosophy is that players should be encouraged to explore, not

punished for difficulty. A player who struggles with specific levels can move forward after completing the majority, then return later with better towers, higher-level heroes, and more experience to tackle the challenges they skipped. The three-star system provides the difficulty ceiling for dedicated players, while the one-star unlock provides the accessible floor for casual players.

9.2 Star System

Each level offers three stars, representing three tiers of completion challenge. The first star is earned by surviving all enemy waves without losing all lives. The second star requires completing the level while retaining a minimum percentage of lives (typically 50% or more). The third star requires completing the level under an additional constraint, which varies by level and may include time limits, resource caps, tower count limits, or no-upgrade restrictions.

Each star grants a one-time SAKART reward. The reward amounts scale with the difficulty of the era: early era stars grant fewer tokens, while later era stars grant progressively more. This scaling reflects both the increased difficulty of later content and the economic design goal of distributing tokens over the long term rather than front-loading rewards in early content. Once a star is earned, replaying the level generates zero additional SAKART, eliminating replay farming as a concern entirely.

10. Scrolls and Mastery

Scrolls are consumable items that provide powerful temporary effects during gameplay. They are the primary tactical resource that players manage during a level, alongside tower placement and hero ability usage. Scrolls are single-use: activating a scroll consumes it permanently, and players must acquire new scrolls to replenish their inventory. Each player account can hold a maximum of 50 scrolls at any time, preventing hoarding and encouraging strategic use.

Scrolls are divided into two categories: standard scrolls and mastery scrolls. Standard scrolls provide gameplay effects during levels (damage boosts, resource generation, crowd control, healing, etc.) and are earned through gameplay progression or purchased with SAKART tokens. Mastery scrolls provide permanent passive bonuses to specific tower types or heroes and are represented as ERC-721 NFTs on the blockchain, making them tradeable on the marketplace.

10.1 Standard Scrolls

Standard scrolls come in several types, each providing a different tactical effect. Damage scrolls boost tower damage output for a limited duration. Economy scrolls generate bonus in-game resources. Crowd control scrolls freeze or slow all enemies on the map. Healing scrolls restore HP to damaged towers. Shield scrolls grant temporary invulnerability to a targeted tower or hero. Each scroll type has multiple tiers of potency, with higher tiers providing stronger effects.

Standard scrolls are consumed when activated during a level. They are not blockchain assets and cannot be traded between players. This design keeps the tactical resource economy simple and predictable. Standard scrolls can be earned as rewards for completing levels with 3 stars, purchased from an in-game shop using in-game currency (not SAKART), or occasionally received as daily login bonuses. The scroll inventory cap of 50 ensures that players must make meaningful choices about which scrolls to bring into each level.

10.2 Mastery Scrolls (ERC-721)

Mastery scrolls are NFT-based permanent passive bonuses. Unlike standard scrolls, mastery scrolls are not consumed during gameplay. Instead, they are activated in the player's loadout before a level begins and provide their bonus for the entire level duration. Mastery scrolls are type-based, not unique: there is no limit on how many copies of a specific mastery scroll can exist, but each player can only activate each type of mastery scroll once per level, with a maximum stack of 3 copies of the same type providing increasing bonuses.

The stack system works as follows: activating 1 copy of a mastery scroll provides a 3% bonus to the relevant stat, 2 copies provide 6%, and 3 copies provide the maximum of 9%. This cap ensures that mastery scrolls enhance gameplay without breaking balance. For example, a player with 3 copies of the "Archer Tower Mastery" scroll would have their Archer Towers deal 9% more damage for the duration of any level where the scroll is activated. Mastery scrolls are tradeable on the marketplace, creating a player-driven economy for these permanent upgrades.

10.3 SAKART Boost Scroll

As confirmed during design review, there is NO SAKART Boost scroll. The token economy is driven entirely by star-based earning, and no consumable item should exist that accelerates or modifies SAKART rewards. This prevents economic exploits and keeps the token distribution system clean and predictable.

11. Skins and Binding

Cosmetic skins are the primary visual customization system and the main use case for the SAKART token economy. Skins change the visual appearance of towers and heroes without affecting gameplay mechanics. They are implemented as ERC-1155 tokens on the Immutable zkEVM blockchain, enabling frictionless trading on the marketplace. Skins are purely cosmetic: a tower with a skin has identical stats and abilities to the same tower without a skin.

The skin system is designed around the era structure. Each tower type has one default skin per era (automatically applied when playing levels in that era) and multiple premium skins that players can acquire through gameplay rewards or marketplace purchases. Premium skins feature more detailed art, special effects, and animation variations that make them visually distinct from the default era skins.

11.1 Skin Binding

When a player equips a skin to a tower, the skin becomes bound to that specific tower instance. Bound skins cannot be traded on the marketplace until they are unbound, which costs a small amount of SAKART. This binding mechanism serves two purposes: it creates a SAKART token sink (burn mechanism) through the bind/unbind fee, and it prevents marketplace speculation on skins that are actively being used in gameplay.

The bind/unbind fee is set low enough to not discourage players from experimenting with different skins, but high enough to create meaningful economic friction against rapid marketplace trading. The exact fee amount is subject to balance testing and may vary based on the skin's rarity tier. Common skins have minimal bind/unbind fees, while rare and legendary skins have higher fees, reflecting their greater marketplace value.

11.2 Marketplace Tooltip

Every item on the marketplace displays a clear tooltip showing: what the item does, which era/level it applies to, its current bind status, and the cost to unbind if applicable. This transparency ensures that players can make informed purchasing decisions without needing to consult external resources. The tooltip system is especially important for mastery scrolls, where the stack bonus progression (3%, 6%, 9%) must be clearly communicated.

12. Player vs. Player (PvP)

PvP in Sakartvelo Defenders is an asynchronous attack-then-defend format designed for fairness and accessibility. Unlike synchronous PvP tower defense, which requires both players to be online simultaneously, the asynchronous format allows players to attack an opponent's preset defense at their convenience, and defend against incoming attacks with AI-managed towers. This design respects the player's time while maintaining competitive depth.

12.1 Attack-Then-Defend Format

Each PvP engagement consists of two phases. In the Attack phase, the player selects a league-level map and sends enemy waves against the opponent's pre-built defense. In the Defend phase, the player's own defense is tested against waves sent by their opponent. The winner is determined by total score across both phases, where score is based on lives remaining, time survived, and enemies defeated.

The turn order for who attacks first is determined by a Rock-Paper-Scissors (RPS) mechanic at the start of each engagement. Both players secretly choose their throw, and the winner attacks first. Attacking first provides a slight psychological advantage (knowing the defense layout) but also reveals information that the defender can use in the second phase. This simple RPS mechanic adds a layer of pre-game strategy without introducing complexity that would alienate casual players.

12.2 Loadout System

PvP uses identical loadouts for both players to ensure absolute fairness. When a PvP engagement begins, both players are assigned the same tower roster, hero, and scroll selection. This eliminates pay-to-win concerns, as no player can gain an advantage through superior NFTs or token spending. The skill expression in PvP comes entirely from tower placement strategy, timing of hero abilities, and scroll usage.

There are no re-rolls on scrolls in PvP. Each player receives a fixed scroll selection at the start of the engagement and must make the best use of what they are given. This constraint encourages strategic planning and rewards players who can adapt their tactics to available resources rather than relying on lucky draws.

12.3 Defense Setup

Players build their defense on a standardized PvP map layout before entering matchmaking. The defense setup uses pre-built tower configurations that are executed by AI when the player is being attacked. The AI uses a predefined priority system: towers target the enemy closest to the exit, use abilities when enough enemies are in range, and do not make placement changes during the defense. This ensures that the defense performs consistently regardless of whether the player is online.

The PvP league system is organized by skill tiers rather than eras. PvP should not have era restrictions; only league levels should gate matchmaking. This means a player who has completed Era 5 can compete in PvP against players at the same league level regardless of their campaign progress. The league system uses a standard ELO-based rating with promotion and demotion between tiers.

12.4 Hero and Scroll Selection

For the launch version, PvP uses a fixed rotating roster of heroes and scrolls that changes weekly. This design ensures balance without requiring continuous meta adjustments, and it keeps the PvP experience fresh by regularly changing the available strategic options. The rotating roster is selected by the development team based on balance testing, with community feedback incorporated through regular surveys. Future versions may introduce a draft-pick system where players alternately select heroes and scrolls from a shared pool.

13. Survival Mode

Survival Mode is an endless wave defense challenge separate from the main campaign. Players select a map and defend against increasingly difficult waves that continue until all lives are lost. There is no win condition in Survival Mode; the goal is to survive as many waves as possible and achieve the highest score on the global leaderboard. Survival Mode is the primary competitive format for players who have completed the campaign and seek ongoing challenges.

Survival Mode uses the player's full tower roster, hero collection, and scroll inventory from the campaign. Unlike PvP, there are no loadout restrictions, allowing players to use their best available assets. However, the scaling difficulty ensures that even the most powerful loadout will eventually be overwhelmed, typically around wave 50–100 depending on player skill. The global leaderboard tracks the highest wave reached, total enemies defeated, and score, with seasonal resets that award SAKART tokens to top performers.

14. Seasons

Seasons are quarterly content updates that introduce new gameplay features, limited-time events, and exclusive cosmetic rewards. Each season lasts approximately three months and is themed around a specific aspect of Georgian culture or history that may not fit into the main chronological campaign. For example, a season might focus on Georgian mythology (featuring folklore-based enemies and towers), Georgian cuisine (with food-themed cosmetic skins), or specific historical events like the Battle of Didgori.

Seasons serve several design purposes. They provide ongoing content for players who have completed the campaign, creating reasons to return regularly. They introduce variety into the gameplay experience by temporarily modifying core rules (such as double damage for a specific tower type, or altered enemy compositions). They generate marketplace activity through limited-edition cosmetic skins that are only available during the season. And they provide additional SAKART burn mechanisms through season pass purchases and exclusive item acquisition.

Each season includes a free track with basic rewards available to all players, and a premium track purchased with SAKART that offers enhanced rewards including exclusive skins, scroll packs, and mastery scroll fragments. The season pass price is set to create a meaningful but not punitive SAKART sink, contributing to the target annual deflation rate.

15. Endgame

The endgame begins when a player completes all 200 campaign levels and has access to the full tower roster, hero collection, and competitive modes. The endgame is designed to provide long-term engagement through three pillars: mastery (perfecting three-star completions and Survival Mode performance), collection (acquiring rare skins and mastery scrolls through the marketplace), and competition (climbing PvP and Survival leaderboards).

Mastery challenges include achieving three stars on all 200 levels, completing levels under self-imposed restrictions (no upgrades, single tower type only, hero-only runs), and setting high scores in Survival Mode. These challenges provide bragging rights and are tracked on the player's public profile. Collection goals revolve around acquiring the full set of cosmetic skins for each tower type and hero, including rare limited-edition skins from past seasons. Competition is maintained through seasonal leaderboard resets and regular PvP tournaments with SAKART prize pools.

The endgame economy is critical for long-term token health. With campaign star rewards largely earned by this point, the endgame introduces additional SAKART sinks through cosmetic purchases, season passes, and tournament entry fees. The balance between endgame sinks and remaining campaign earning ensures that the token economy remains healthy without requiring players to spend real money to continue enjoying the game.

16. Educational System

The educational system is one of the three foundational pillars of Sakartvelo Defenders. The game is designed to teach Georgian history organically through gameplay, with educational content woven into the fabric of the experience rather than presented as separate lessons. The educational system operates on three levels: passive learning through visual design and environmental storytelling, active learning through historical pop-ups between levels, and deep learning through an optional in-game encyclopedia.

Passive learning happens naturally as players progress through the campaign. The era-specific art styles, tower designs, enemy types, and map environments all communicate historical information visually. A player who spends 20 levels defending against Mongol armies in Era 4 will develop an intuitive understanding of the Mongol period's impact on Georgia, even if they never read a single educational pop-up. The visual design language is the most powerful educational tool in the game.

Historical pop-ups appear between every level and provide 2–4 sentences of context about the events, figures, or locations featured in the upcoming level. These pop-ups are skippable for players who want to focus on gameplay, but they are designed to be engaging enough that most players will read them. Historical pop-ups do NOT cost SAKART to read. Making education paywalled would undermine the game's core philosophy of cultural preservation and accessibility.

16.1 Cultural Footer

The game includes a cultural footer on the main menu and loading screens that displays rotating Georgian cultural facts, proverbs, and historical quotes. This passive educational element adds flavor without interrupting gameplay and reinforces the game's identity as a cultural product. The cultural footer content is curated to be interesting and diverse, covering topics including:

- Georgian wine-making traditions — the world's oldest, with 8,000-year-old archaeological evidence of wine production at Gadachrili Gora and

Shulaveris Gora, using the traditional qvevri (clay vessel) method that is itself recognized as UNESCO Intangible Cultural Heritage

- The Georgian alphabet — comprising three writing systems (Mrgvlovani, Nuskhuri, and Mkhedruli) inscribed together as UNESCO Intangible Cultural Heritage in 2016, and one of only about 14 independently created writing systems in the world
- Georgian polyphonic singing — recognized by UNESCO as Intangible Cultural Heritage in 2001 and included on the Voyager Golden Record, representing one of humanity's greatest musical achievements
- The supra tradition — the Georgian feast with its formal toasting hierarchy led by the tamada (toastmaster), which serves as a social institution central to Georgian cultural identity
- Georgian proverbs and wisdom sayings that illuminate the national character

16.2 Historical Accuracy Policy

The game is committed to historical accuracy wherever possible. All historical events, figures, and cultural elements depicted in the game are based on accepted historical scholarship and Georgian historical sources. Where the historical record is ambiguous or contested, the game presents the most widely accepted interpretation and notes the existence of alternative views. The Stalin era (Era 8) is handled with particular care, presenting factual information about his actions and their consequences without glorification or erasure.

The historical accuracy policy extends to visual design. Tower skins, enemy designs, map environments, and hero appearances are all based on historical research and archaeological evidence. The Art Style Guide provides detailed reference material for each era's visual language, ensuring that the AI-generated art assets maintain consistency with historical reality. Anachronistic elements are avoided unless they serve a specific gameplay purpose and are clearly identified as non-historical.

16.3 Verified Sources and Methodology

All historical claims in this Bible have been verified against authoritative sources including: the "Kartlis Tskhovreba" (Life of Kartli) medieval Georgian chronicles; English-language academic scholarship on Georgian history; Wikipedia articles cross-referenced with primary sources; UNESCO records for cultural heritage claims; and web-searched scholarly databases. Where multiple sources disagree (such as the numbers at the Battle of Didgori), the Bible presents both the traditional Georgian account and the modern scholarly assessment, allowing players to understand that history is not always settled.

17. Mythology Integration

Georgian mythology is deeply intertwined with the country's historical identity, and Sakartvelo Defenders incorporates mythological elements as complementary to the historical narrative rather than as a separate system. Mythological creatures and heroes appear primarily in Era 0 (Ancient Colchis, where the boundary between myth and history is naturally blurred) and in special seasonal events that explore specific aspects of Georgian folklore.

Key mythological elements include:

- **Devi** — Many-headed ogres whose heads can regenerate if any of them are cut off (comparable to the Lernaean Hydra). Devi are the most gameplay-relevant mythological creatures in the game, as their head-regeneration mechanic translates directly into a boss encounter where players must completely destroy the devi before its heads regrow. Bakbak-devi is a specific variant often portrayed as a multi-headed, shrewd, man-eating ogre.
- **Kaji** — A race of evil spirits, often portrayed as magic-wielding, demonic metalworkers. They lived in Kajeti (Kajaveti) and used their magical powers against humans. In the game, kaji are framed as captured or bound spirits whose magical forging abilities are harnessed by the player for tower upgrades in the early eras, rather than as friendly blacksmiths. The educational content clarifies that kaji are traditionally malevolent beings in Georgian folklore.
- **Amirani** — The Georgian Prometheus figure. Amirani was a legendary hero who dared to challenge the gods and stole fire for humanity. As punishment, he was chained to a mountain in the Caucasus (some versions specify Mount Kazbek), where a raven or eagle pecks at his liver daily, only for it to regenerate each night. The Amirani myth is central to Georgian cultural identity and represents the theme of defiance against impossible odds — a perfect thematic match for a tower defense game. Amirani could feature as a seasonal event hero with a "Stolen Fire" ultimate ability that temporarily grants all towers maximum damage.
- **The Golden Fleece** — The central mythological narrative of Era 0, reflecting the historical reality of Colchian gold mining and the ancient technique of gold-panning using sheep fleeces to trap gold particles in mountain streams.
- **Borjgali** — The ancient Georgian solar symbol, resembling a stylized sun with seven rays. The Borjgali has been a national emblem for millennia and appears on the modern flag of Georgia. In the game, the Borjgali could be incorporated into the visual design language as a recurring motif

in tower ornamentation, hero ability effects, and the game's logo. It represents eternal motion and the sun's life-giving power.

- **Pre-Christian deities** — Armazi (supreme sky/thunder god), Zaden (fertility god), Gatsi (moon-associated), and Gaim (sun-associated) provide rich naming opportunities for early-era towers, abilities, and map features. The Armazi fortress near Mtskheta, which housed the great idol of Armazi, is already a map location in Era 1.

The game also draws on the epic poem "The Knight in the Panther's Skin" by Shota Rustaveli for heroic inspiration, with characters and themes from the poem influencing hero designs and ability concepts. The identity of the animal in the title is debated among scholars: in modern Georgian the word refers to a tiger, but in Rustaveli's time it could have meant panther or leopard. The game uses "The Knight in the Panther's Skin" as the primary English title, with the scholarly debate noted in educational content.

The mythology system is designed to enhance rather than replace the historical content. Players who are interested in mythology will find rich references throughout the game, while players who prefer pure history can focus on the historical eras without missing any core gameplay content. The distinction between mythological and historical content is always clearly communicated through the educational system, ensuring that players understand which elements are based on historical records and which are drawn from folklore.

18. SAKART Token Economy

SAKART is the ERC-20 utility token that powers the Sakartvelo Defenders economy. The token has a fixed hard cap of 1 billion (1,000,000,000) SAKART, with zero pre-mine and zero team allocation. All tokens enter circulation exclusively through player gameplay: stars earned on campaign levels are the sole distribution mechanism. This design ensures that the token economy is entirely player-driven, with no inflation from team unlocks, investor allocations, or algorithmic minting.

18.1 Token Distribution

Token distribution follows a star-based system. Each of the 200 campaign levels offers 3 stars (600 total stars). Each star, when earned for the first time, grants a one-time SAKART reward. The reward per star scales with the era, with early eras granting smaller rewards and later eras granting larger ones. The total token supply allocated to campaign rewards is approximately 700 million

SAKART (70% of the hard cap), with the remaining 300 million reserved for seasonal rewards, competitive prizes, and marketplace liquidity incentives.

Because each star can only be earned once per player, the total number of SAKART tokens in circulation is directly proportional to the number of active players and their campaign progress. A player who earns all 600 stars across the full campaign will have accumulated a substantial but finite amount of SAKART. There is no mechanism for earning SAKART through replay, grinding, or any activity other than first-time star completions. This structural design makes replay farming mathematically impossible.

18.2 Burn Mechanisms

The SAKART economy includes multiple burn mechanisms designed to create sustained deflationary pressure. The primary burn mechanisms are: skin purchases on the marketplace (marketplace fees are partially burned), skin bind/unbind fees, season pass purchases, and cosmetic effect unlocks. Additional burn mechanisms under consideration include scroll reroll fees (allowing players to reroll their scroll selection for a level at a SAKART cost), cosmetic tower effects (visual effects applied to towers that cost SAKART per activation), and leaderboard entry fees for competitive tournaments.

The target annual deflation rate is 2–5% of circulating supply, achieved through the combination of all burn mechanisms. This deflation is not aggressive enough to create hoarding behavior (which would harm the gameplay economy) but sufficient to offset the slow trickle of new tokens entering circulation as new players join and earn stars. The fee structure is designed to be invisible to casual players while providing meaningful economic activity for engaged players.

18.3 Anti-Inflation Design

The SAKART economy is structurally anti-inflationary due to several design choices. First, the star-based earning system caps total possible earnings per player. Second, the absence of replay farming eliminates the most common Web3 game inflation vector. Third, the fixed hard cap of 1 billion tokens provides an absolute ceiling on supply. Fourth, the multiple burn mechanisms create continuous deflationary pressure. Fifth, the gameplay economy (in-game resources for tower placement and upgrades) is entirely separate from the SAKART economy, preventing token inflation from affecting gameplay balance.

19. Fees and Revenue Model

The revenue model for Sakartvelo Defenders is built around a transparent transaction fee applied to all marketplace trades. The total fee is 1.8% per transaction, distributed as follows: 1.3% to the development treasury for ongoing game development, server costs (minimal due to PWA architecture), and team compensation; 0.5% automatically compounded into the marketplace liquidity pool to ensure healthy trading conditions. No additional fees are charged for gameplay, account creation, or any other activity.

The 1.8% fee structure is designed to be low enough to encourage active marketplace trading while generating sufficient revenue to sustain development. As a solo developer with a $100/month budget, the minimal server costs (PWA has zero server hosting) mean that marketplace fees are almost entirely profit. The automatic LP compounding (0.5%) ensures that the marketplace remains liquid even during low-activity periods, preventing the common Web3 problem of illiquid markets that discourage trading.

19.1 Treasury and Developer Withdrawal

The development treasury accumulates the 1.3% developer fee from all marketplace transactions. Withdrawals from the treasury are subject to a monthly cap to prevent sudden large withdrawals that could harm community trust. The monthly cap is set to cover development costs plus a reasonable developer salary, with any excess remaining in the treasury for future development needs. Treasury activity is fully transparent and auditable on-chain.

The withdrawal mechanism requires a timelock: withdrawal requests are queued for 48 hours before execution, during which they are publicly visible on the blockchain. This transparency mechanism ensures that the community can verify that treasury funds are being used for legitimate development purposes. As a solo developer, this level of transparency is both a business practice and a competitive advantage, building trust with the player community that the game will be maintained and improved over the long term.

20. Smart Contracts

Sakartvelo Defenders deploys on the Immutable zkEVM Layer 2 network, which provides zero gas fee transactions for players while maintaining the security guarantees of Ethereum. The smart contract architecture consists of three primary contracts: the SAKART token contract (ERC-20), the Skin contract

(ERC-1155), and the Mastery Scroll contract (ERC-721). All contracts follow established security best practices and will be audited before mainnet deployment.

20.1 SAKART Token Contract (ERC-20)

The SAKART token contract implements the standard ERC-20 interface with custom modifications for the game's requirements. The total supply is fixed at 1 billion tokens with no minting function. Token transfers to and from the game's designated contracts (marketplace, season pass, bind/unbind) are gasless through Immutable's account abstraction. The contract includes a blacklist function for fraud prevention and multi-account detection, governed by a multisig wallet controlled by the developer.

20.2 Skin Contract (ERC-1155)

The ERC-1155 standard is used for cosmetic skins because it supports both fungible and non-fungible tokens within a single contract. Common skins are semi-fungible (multiple identical copies exist), while rare and legendary skins are non-fungible (unique or limited edition). The contract includes metadata URI pointers to decentralized storage (IPFS) for skin artwork and description data, ensuring that skin assets remain accessible even if the game's servers go offline.

20.3 Mastery Scroll Contract (ERC-721)

Mastery scrolls use the ERC-721 standard because each scroll is a distinct non-fungible asset, even when multiple scrolls of the same type exist. The contract tracks scroll type, stack count, and activation status. When a player activates a mastery scroll for a level, the contract verifies ownership and stack count before applying the bonus. The contract also manages the bind/unbind mechanism, requiring a SAKART payment for unbinding that is automatically burned.

21. Blockchain Integration

The blockchain integration follows the "invisible blockchain" design philosophy: players should never feel like they are interacting with a blockchain product unless they choose to engage with marketplace or token features. All core gameplay, including the full 200-level campaign, PvP, Survival Mode, and hero progression, functions identically whether or not the player has connected a wallet. Blockchain features enhance the experience for players who want true asset ownership and tradable items, but they are never required for gameplay.

21.1 Wallet Integration

Wallet connection is handled through Immutable's Passport wallet abstraction system. Passport provides a seamless onboarding experience that allows players to create a wallet using social login (email, Google, Apple) without needing to understand blockchain concepts. For players who already have Web3 wallets, Passport supports direct connection through MetaMask, WalletConnect, and other popular wallet providers. The wallet connection is optional and can be deferred indefinitely without affecting gameplay access.

21.2 Chain Outage Fallback

The game implements a chain outage fallback system. If the Immutable zkEVM network experiences downtime, gameplay continues uninterrupted. All gameplay state (level progress, star completions, hero experience) is tracked locally and synced to the blockchain when connectivity resumes. Blockchain-specific transactions (marketplace trades, token transfers, NFT operations) are queued locally and submitted automatically in order when the chain comes back online. This ensures that players never lose progress due to infrastructure issues.

21.3 Invisible Blockchain Philosophy

The invisible blockchain philosophy extends to every aspect of the game's design. There is no blockchain tutorial, no crypto jargon in the UI, and no requirement to understand gas, wallets, or smart contracts to enjoy the full gameplay experience. The UI uses standard gaming terminology ("Shop" instead of "Marketplace," "Items" instead of "NFTs," "Coins" instead of "Tokens") for all core gameplay interactions. Blockchain-specific terminology is only used in the dedicated marketplace section, which players only see if they choose to explore trading features.

22. Multi-Account Prevention

Multi-account prevention is critical for the SAKART economy because each account can earn SAKART through star completions. Without prevention, a single person could create multiple accounts, complete the early (easier) levels on each account to farm SAKART, and then consolidate the tokens through marketplace trades. This would undermine the structural anti-inflation design and devalue the token for legitimate players.

The prevention system uses multiple layers. At the wallet level, Passport's identity system ties each account to a verified identity (email, phone, or social login), making it difficult to create large numbers of accounts. At the gameplay level, progression requirements (15/20 stars to unlock the next era) ensure that farming accounts must invest significant time before earning meaningful SAKART. At the transaction level, the marketplace monitors for patterns consistent with multi-account farming (multiple accounts sending tokens to a single address within a short timeframe) and flags suspicious activity for investigation.

The blacklist function in the SAKART token contract allows flagged accounts to be suspended from token transactions while preserving their gameplay progress. This ensures that legitimate players who happen to share IP addresses (family members, university networks) are not unfairly penalized. The threshold for blacklisting is intentionally conservative, requiring multiple indicators of farming behavior before action is taken.

23. Competitive Analysis

23.1 Web2 Tower Defense Landscape

The Web2 tower defense market is dominated by Kingdom Rush, Bloons TD 6, and Arknights. These games represent the gold standard for tower defense gameplay and provide the benchmark for quality that Sakartvelo Defenders must meet. Kingdom Rush is known for its tight level design and tower upgrade system. Bloons TD 6 offers enormous content depth with hundreds of upgrades and game modes. Arknights combines tower defense with gacha collection mechanics and a rich narrative.

Sakartvelo Defenders borrows the best elements from each: Kingdom Rush's focused level design and upgrade clarity, Bloons TD 6's content volume and replayability, and Arknights' narrative engagement (adapted to historical rather than fictional storytelling). The key differentiator is the historical education component, which provides a unique value proposition that none of these competitors offer. Additionally, the invisible blockchain integration provides true asset ownership without the friction that has prevented Web3 games from competing with Web2 quality.

23.2 Web3 Gaming Landscape

Web3 tower defense games are a nascent category with few established titles. Most Web3 games in this space have failed because they prioritize tokenomics

over gameplay, resulting in products that are essentially financial instruments with game-like interfaces. Sakartvelo Defenders takes the opposite approach: the gameplay is designed to be competitive with Web2 titles, and blockchain is an invisible infrastructure layer that enhances the experience without defining it.

The invisible blockchain philosophy is the primary competitive advantage in the Web3 space. By eliminating wallet requirements, gas fees, and crypto jargon from the core experience, Sakartvelo Defenders can attract the much larger Web2 gaming audience while still providing Web3 benefits (true ownership, player-driven economy, transparent governance) to players who choose to engage with those features. This bridge between Web2 and Web3 audiences is the game's most significant market opportunity.

24. Art Pipeline

24.1 Style Bible System

The Art Style Guide provides the authoritative reference for all visual assets. The guide defines a 2D isometric art style with cel-shaded lighting, a consistent character proportion system (10-unit body scale), era-specific color palettes (8 colors per era), and tower silhouette guidelines that ensure each tower type is visually distinct at small sizes. Every AI generation prompt includes era-appropriate keywords drawn from the style guide, ensuring that assets from different eras are visually cohesive within their period while being clearly distinct across periods.

24.2 AI Generation Workflow

The workflow begins with a prompt template that incorporates the style guide parameters for the target era and asset type. The prompt includes era-specific color palette references, architectural keywords (e.g., "Svan medieval stone tower" for Era 4, "Soviet brutalist concrete" for Era 8, "Colchian bronze-age wooden fortification" for Era 0), and composition guidelines (isometric perspective, cel-shaded lighting, no text overlays). Generated images are reviewed against the style guide consistency checklist, which verifies color accuracy, proportion correctness, silhouette clarity, and era-appropriate architectural details.

25. Tech Stack

Component	Technology	Rationale
Game Engine	Godot 4.x	Free, open-source, excellent 2D support, PWA export
Deployment	PWA (Progressive Web App)	Zero server costs, offline play, browser-based
Blockchain	Immutable zkEVM	Zero gas, EVM compatible, Passport wallet
Token Standard	ERC-20 (SAKART)	Well-established, wide tooling support
Skin Standard	ERC-1155	Fungible + non-fungible in one contract
Scroll Standard	ERC-721	Distinct NFT assets with metadata
AI Art	Stable Diffusion / DALL-E	Style guide-driven prompt architecture
Asset Storage	IPFS	Decentralized, persistent, censorship-resistant
Website	Vercel (static)	Free tier, fast CDN, automatic HTTPS
Budget	$100/month	Vercel free + Immutable free + domain only

25.1 Godot 4.x and PWA Architecture

Godot 4.x was selected as the game engine for several reasons: it is free and open-source, it has excellent 2D rendering capabilities, it supports PWA export natively, and it has a small but active community. The PWA deployment model means the game runs entirely in the browser with no server-side game logic. All gameplay computation happens on the player's device, eliminating the need for game servers. The only server interactions are blockchain transactions (handled by Immutable's infrastructure) and website hosting (handled by Vercel's free tier).

The PWA architecture also provides offline gameplay capabilities. The full campaign, all tower and hero data, and all map assets are cached locally through the PWA service worker. Players can continue playing the campaign even without internet access, with blockchain transactions queued for later submission. This

offline capability is a significant competitive advantage over cloud-dependent games and aligns with the design philosophy of removing all friction from the player experience.

25.2 Blockchain Infrastructure

Immutable zkEVM provides the blockchain infrastructure with two critical advantages: zero gas fees for players and EVM compatibility. Zero gas is essential for the invisible blockchain philosophy, as any visible transaction cost would immediately break the illusion of a traditional game. EVM compatibility ensures that the smart contracts can use established standards (ERC-20, ERC-721, ERC-1155) and benefit from the extensive Ethereum tooling ecosystem.

26. Design Principles

The following principles are the non-negotiable rules that govern all design decisions in Sakartvelo Defenders. When any design question arises, these principles provide the definitive answer. They are not suggestions or guidelines; they are hard constraints that cannot be overridden without a deliberate design review.

26.1 No-Nerf Rule

Once a tower, hero, scroll, or ability is released, its core functionality will never be nerfed (reduced in power). If something is too powerful, the response is to buff everything else to match, not to weaken the overperforming element. This principle protects player investment and builds trust.

26.2 Invisible Blockchain

No player should ever feel like they are interacting with a blockchain product unless they choose to engage with marketplace or token features. All UI language uses standard gaming terminology. No blockchain tutorial is required. No gas fees are visible to players.

26.3 Historical Accuracy

The game is chronologically accurate and committed to truthful representation of Georgian history. Eras follow the correct historical sequence. Towers, enemies, and heroes are based on real historical references. When the historical record is

ambiguous, the game acknowledges the uncertainty rather than presenting speculation as fact. All dates, figures, and events have been verified against authoritative academic sources.

26.4 Solo Developer Feasibility

Every feature must be implementable by a solo developer on a $100/month budget. This constraint drives technology choices, content scope, and feature prioritization.

26.5 Accessibility

The game must be accessible to players with no prior experience with tower defense games, blockchain technology, or Georgian history. The tutorial system teaches core mechanics organically. Difficulty is calibrated to allow casual players to progress while providing meaningful challenges for dedicated players. Educational content is available but never forced. The blockchain is invisible to players who do not seek it out.

27. Roadmap

Phase	Timeline	Key Deliverables	Budget
Phase 1: Foundation	Months 1–3	Core gameplay loop, Era 0–1 (40 levels), basic UI, Godot PWA setup	$0 (free tools)
Phase 2: Expansion	Months 4–6	Era 2–4 (60 more levels), hero system, scroll system, synergy system	$0 (free tools)
Phase 3: Blockchain	Months 7–9	Immutable integration, SAKART contract, ERC-1155 skins, marketplace MVP	$0 (Immutable free tier)
Phase 4: Full Campaign	Months 10–14	Era 5–9 (100 more levels), all 10 tower types, PvP system, Survival Mode	$0 (free tools)
Phase 5: Polish	Months 15–18	Art refinement, educational system, seasons, competitive features	$300 (domain + marketing)
Phase 6: Launch	Month 19	Full game launch, marketing push, community building, marketplace open	$100/month ongoing

28. Changelog (v3.0 to v4.0)

This section documents all changes made to the Game Design Bible between version 3.0 and version 4.0. The changes are driven by a comprehensive historical accuracy review that identified five factual errors, eight nuanced claims requiring clarification, and significant previously omitted content. The v4.0 update represents the most thorough historical revision in the document's history.

Section	Change	Reason
3.1 (Era 0)	Date range changed from ~1500 BC – ~300 BC to ~600 BC – ~100 BC	Kingdom of Colchis proper dates to 6th century BC; earlier period was pre-kingdom
3.2 (Era 1)	Added pre-Christian deities (Armazi, Zaden, Gatsi, Gaim)	Important cultural context for pre-Christian Iberia
3.2 (Era 1)	Clarified Gorgasali's founding of Tbilisi	He developed existing settlement, not built from nothing
3.3 (Era 2)	Added distinction between 645 AD Arab conquest and 736 CE Emirate establishment	These are distinct events separated by nearly a century
3.3 (Era 2)	Added Bagrat III the Unifier	Essential figure who unified Georgia in 1008
3.3 (Era 2)	Added Didi Turkoba (Great Turkish Onslaught)	Essential context for understanding the Golden Age
3.4 (Era 3)	Added Tamar's two marriages (Yuri Bogolyubsky and David Soslan)	Significant historical detail
3.4 (Era 3)	Renamed Mkhargrdzeli to Zakarian; acknowledged Armenian origin	Standard academic name is Zakarian
3.4 (Era 3)	Added Empire of Trebizond as Georgian client state	Demonstrates extent of Georgian power
3.4 (Era 3)	Added Didgori numbers dispute	Historical nuance about source reliability
3.5 (Era 4)	Added Queen Rusudan and George V the Brilliant (two reigns)	Essential figures of the Mongol era
3.5 (Era 4)		

	Changed boss from Subutai's proxy to General Chormaqan	Chormaqan led the actual conquest; Subutai was reconnaissance
3.6 (Era 5)	Added Vakhtang VI the Lawgiver	Major figure: legal code, scholar-king
3.6 (Era 5)	Added Totleben betrayal at Aspindza (1770)	Foreshadows Georgievsk betrayal
3.6 (Era 5)	Changed Erekle II ability to reflect tragic diplomacy	His diplomacy ultimately failed; more historically accurate
3.7 (Era 6)	Removed fabricated name "Gia Abesalomashvili"	No historical record of this person exists
3.7 (Era 6)	Added Prince Pyotr Bagration as hero	Authentic Georgian-born Russian military hero
3.8 (Era 7)	Added Transcaucasian Federation context	Important precursor to Georgian independence
3.8 (Era 7)	Added British military presence 1918–1920	Historical context for DRG's brief existence
3.8 (Era 7)	Clarified League of Nations: application postponed, not approved	Accurate representation of international recognition
3.9 (Era 8)	Corrected WWII casualties from 350,000 to ~190,000 KIA	Authoritative sources confirm ~190,000 killed in action
3.9 (Era 8)	Added 1956 Tbilisi Massacre	Essential event for understanding Stalin legacy
3.10 (Era 9)	Added 2023–2024 political crisis and frozen EU candidacy	Most current historical context
3.10 (Era 9)	Clarified 2008 war was with Russian military primarily	Russia was the primary combatant, not just separatists
5.2 (Bosses)	Changed Mongol boss to Chormaqan; added Timur	Historically accurate Mongol commanders
6 (Heroes)	Added Bagrat III, Rusudan, George V, Vakhtang VI, Merab Kostava	Essential historical figures previously omitted
6 (Heroes)	Changed Rustaveli ability to Cultural Inspiration	Poet, not military commander
16.1 (Footer)		Significant Georgian cultural heritage elements

	Added polyphonic singing, Borjgali, qvevri wine, three writing systems	
17 (Mythology)	Added Amirani myth and Borjgali symbol	Core Georgian mythological elements
17 (Mythology)	Clarified devi as many-headed with regeneration mechanic	Distinctive gameplay-relevant feature
17 (Mythology)	Clarified kaji as evil/demonic spirits, not benign blacksmiths	Accurate representation of folklore
17 (Mythology)	Added note on "Panther's Skin" title debate	Scholarly nuance
16.3 (New)	Added Verified Sources and Methodology section	Transparency about historical verification process